

Credible on Paper? Green Banking Policy, Access to Finance, and Institutional Capacity in the Greening of Firms Worldwide

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Abstract

Green banking policy frameworks coordinated through the Sustainable Banking and Finance Network (SBFN) have been adopted across more than sixty emerging and developing economies since 2012, on the premise that supervisory guidelines directing banks to price environmental risk into lending will strengthen the link between firms' access to credit and their green investment behaviour. We test this premise using two independent World Bank Enterprise Survey samples. The primary sample harmonizes the Bank's full 2018–2020 Green Economy Module rollout across 41 economies (28,042 firms) on a rich seven-item green-practice-adoption outcome, matched to each economy's SBFN policy-adoption status and Worldwide Governance Indicators regulatory quality as of its survey year. Because SBFN status is fixed within country in a single-wave cross-section, a classical country-fixed-effects specification cannot identify it or its interactions; we show this explicitly, then re-estimate the relationship with a Bayesian hierarchical model (country-varying slopes) and a causal forest that impose no linear-interaction functional form. All three approaches agree: bank credit access is robustly associated with green practice adoption (average marginal effect of 13 percentage points in

the baseline specification), but neither SBFN policy adoption nor regulatory quality moderates it—both institutional variables instead shift the general level of green adoption directly. We then subject this finding to the sharpest test we can construct: a second, independently assembled sample of 162 further economies (89,797–100,115 firms depending on specification), spanning 2021–2026 and reaching, for the first time in this literature, into high-income economies such as Australia, Germany, and Japan, built from a World Bank standardized cross-country database not previously exploited for this purpose. On a narrower CO2-emissions-monitoring outcome and with country fixed effects the primary sample’s single-wave design cannot support, credit access remains significant while both SBFN and regulatory-quality interactions remain statistically indistinguishable from zero—the identical qualitative pattern, on a different outcome, a different set of economies, and a materially sharper identification strategy. The causal forest’s feature-importance decomposition attributes the bulk of genuine effect heterogeneity in the primary sample (75.6%) to firm size rather than to either institutional moderator (regulatory quality 19.9%; SBFN status 2.6%). One finance-instrument-specific check finds a nominally significant negative credit $\times$ SBFN interaction when finance access is measured via overdraft rather than term credit, though this does not survive a Benjamini–Hochberg correction for multiple comparisons across the full family of interaction tests reported in the paper, and we treat it as suggestive rather than established. We interpret the combined evidence as more consistent with green banking policy’s binding constraint operating through firm-level absorptive capacity than through the country-level regulatory capacity its own institutional design targets, though

we report this as our best current account of an auxiliary, exploratory finding rather than as a settled result. Finally, because none of the above designs exploits exogenous variation in SBFN adoption *timing*, we additionally build a country-year panel (217 economies, 2000–2024) from World Bank World Development Indicators and estimate a Callaway-Sant’Anna staggered event-study around each economy’s own adoption date – the identification strategy this literature most often calls for. It shows no effect on renewable-energy share and a positive association with CO2 emissions per capita that shrinks by roughly 45% once GDP per capita is controlled for and whose event-study profile is more consistent with pre-existing differences in adopting economies’ growth trajectories than with a causal effect of adoption; this independent, differently-identified test converges on the same substantive conclusion as the firm-level analysis.

Keywords: green banking policy, access to finance, institutional complementarity, firm-level green investment, causal forest, hierarchical Bayesian model

JEL: G21, O16, Q56, D22, C11

1. Introduction

Since 2012, more than sixty central banks, banking regulators, and financial-sector regulatory bodies across the emerging and transition world have signed on to a coordinated effort to green their financial systems, adopting national guidelines that direct commercial banks to price environmental and climate risk into lending decisions and to channel credit toward cleaner production. On paper, this represents one of the most ambitious diffusions of

8 environmental financial regulation outside the OECD: economies as differ-
9 ent in income, banking-sector depth, and political regime as Bangladesh,
10 Morocco, Mongolia, and Indonesia now report, to the International Finance
11 Corporation’s Sustainable Banking and Finance Network (SBFN), formal
12 sustainable-finance roadmaps, green-taxonomy pilots, or mandatory environmental-
13 risk disclosure rules for lenders. Whether that adoption changes what actu-
14 ally happens on a factory floor—whether a firm that can get a bank loan is
15 any more likely to install a wastewater treatment system, switch to a more
16 efficient boiler, or measure its own emissions—is a separate and considerably
17 harder question, and it is the one this paper answers.

18 The premise that formal adoption of a policy is not the same as its op-
19 eration in practice is, of course, not new to institutional economics. A large
20 literature on state capacity and policy credibility has long argued that the
21 same law, regulation, or guideline can produce entirely different real-economy
22 outcomes depending on the administrative and judicial infrastructure avail-
23 able to enforce it (La Porta et al., 1998; Acemoglu et al., 2001). Applied to
24 green finance specifically, a parallel body of work has documented that green
25 credit guidelines in China measurably reshape lending allocation and firm
26 environmental performance (Li et al., 2023; Huang et al., 2023; Dai et al.,
27 2025), that firms’ green investment responds to the financing constraints they
28 face (Ghosh and Dutta, 2022; Costa et al., 2024; De Haas et al., 2025), and
29 that the World Bank Enterprise Surveys’ own Green Economy module is a
30 workable lens on firm-level environmental behaviour across dozens of coun-
31 tries at once (Asif et al., 2025; Phan, 2026). What is missing is the connection
32 between these two literatures: a direct test, at firm level and across a wide

33 cross-section of the very economies the green-banking-policy movement tar-
34 gets, of whether the credit-to-green-investment channel that these guidelines
35 are designed to activate is actually stronger where the guidelines exist—and
36 whether that depends on the state capacity to make the guideline credible in
37 the first place.

38 This paper closes that gap using firm-level data from two independently
39 assembled samples of World Bank Enterprise Survey (WBES) waves. The
40 primary sample covers 47 country-years fielded between 2018 and 2023 that
41 carry the Bank’s full Green Economy module—a battery of questions on emis-
42 sions monitoring, energy audits, and the adoption of climate-friendly produc-
43 tion measures—alongside the Surveys’ standard access-to-finance module; a
44 second, global sample extends this to 162 further country-years fielded be-
45 tween 2021 and 2026, reaching for the first time in this literature into high-
46 income economies with no SBFN involvement at all (Section 4). We match
47 each economy to its Sustainable Banking and Finance Network policy status
48 as of its survey year, distinguishing genuine ex-ante policy adopters from
49 economies that joined only afterward, and to the World Bank’s Worldwide
50 Governance Indicators as a measure of the regulatory state capacity avail-
51 able to make that policy credible. Econometrically, we depart deliberately
52 from the pooled fixed-effects specification that has become the default in
53 this literature. Because SBFN status is fixed within country in a single-wave
54 cross-section, a country fixed effect absorbs it entirely, and a naive interaction
55 term estimated by pooled logit—which we report first, as a baseline, precisely
56 to make this limitation visible rather than to hide it—returns no detectable
57 policy-moderation effect at all. We show that this null result is not evidence

58 of an absent relationship so much as a symptom of forcing a fundamentally
 59 hierarchical question (does a level-2, country-level policy variable reshape
 60 a level-1, firm-level financing relationship?) into a flat, single-level model
 61 with too few country clusters to lean on for inference (Cameron and Miller,
 62 2015). We then re-estimate the same relationship with tools built for exactly
 63 this structure: a Bayesian hierarchical logistic model with country-varying
 64 slopes (Gelman and Hill, 2007; Gelman et al., 2020), a causal-forest / double-
 65 machine-learning (DML) estimator (Wager and Athey, 2018; Chernozhukov
 66 et al., 2018; Athey et al., 2019; Athey and Imbens, 2019) that lets the treat-
 67 ment effect of credit access vary flexibly across the institutional landscape
 68 rather than through a single estimated coefficient, and, on the much larger
 69 global sample, a country-fixed-effects specification that the primary sample’s
 70 single wave cannot support.

71 This paper contributes to the literature in six respects. First, it pro-
 72 vides what is, to our knowledge, the first direct empirical test of whether
 73 SBFN-style green banking policy adoption changes the firm-level relation-
 74 ship between finance and green investment, rather than only the aggregate
 75 volume of green lending reported by banks themselves—a firm-level, demand-
 76 side complement to the bank-level, supply-side evidence that dominates the
 77 existing green-credit-policy literature. Second, it extends the World Bank
 78 Enterprise Survey’s Green Economy module—used until now mostly within
 79 single countries or single regional waves—into a harmonized 41-country pri-
 80 mary sample spanning Europe, Central Asia, the Caucasus, and the Middle
 81 East and North Africa, giving the paper a breadth of institutional varia-
 82 tion that a single-country or single-region design cannot offer. Third, and

83 unusually for this literature, we subject the primary sample’s central find-
 84 ing to an independently assembled second test: a 162-economy global sam-
 85 ple built from a World Bank standardized cross-country database spanning
 86 2021–2026 that, to our knowledge, no published study of green banking pol-
 87 icy has yet exploited, extending coverage for the first time into high-income
 88 economies entirely outside the SBFN’s own remit and allowing a country-
 89 fixed-effects specification the primary sample’s single-wave design cannot
 90 support. That the same qualitative result—a robust finance-green channel,
 91 no detectable institutional moderation—survives this independent replica-
 92 tion is, we think, considerably more informative than either sample could
 93 be alone. Fourth, methodologically, the paper demonstrates—with the same
 94 underlying question tested twice, on two different samples, under four differ-
 95 ent estimators—exactly where and why a classical fixed-effects approach fails
 96 to detect (or, when country fixed effects are usable, correctly fails to find) an
 97 institutionally-conditioned effect, offering applied researchers working with
 98 similarly structured cross-country firm data a concrete illustration of when
 99 the extra modelling investment is worth making and when a null result is sim-
 100 ply a null result. Fifth, the paper redirects the question of which system-level
 101 margin governs green-credit-policy effectiveness: having ruled out a country-
 102 institutional interaction across four estimators and two samples, we show
 103 (Section 3.4) that the effect heterogeneity that does exist is concentrated at
 104 the level of firm size rather than country institutions, relocating the opera-
 105 tive “system condition” in the Policy Design $\times$ System Conditions $\rightarrow$ System
 106 Outcomes logic this special issue organizes itself around from the country-
 107 regulatory level to the firm-scale level—a mechanism we introduce as an

108 auxiliary, exploratory hypothesis in Section 3.4 and demonstrate empirically
109 in Section 5.4 rather than treat as a pre-registered finding, and are corre-
110 spondingly explicit about the identification limits (Section 4) that a purely
111 cross-sectional design, however extensively replicated, still carries. Sixth, and
112 directly in response to that identification-strategy limit, Section 6 builds an
113 entirely independent country-year panel from World Bank World Develop-
114 ment Indicators and estimates a Callaway-Sant’Anna staggered event-study
115 exploiting SBFN’s own adoption-timing variation—the design class this spe-
116 cial issue’s methodological guidance names first—and finds that it, too, does
117 not support a beneficial institutional effect on aggregate environmental out-
118 comes, converging with the firm-level evidence from a design built specifically
119 to meet an identification bar the firm-level data cannot.

120 The paper also speaks directly to a growing body of recent scholarship in
121 this journal on the institutional and financial determinants of firm and bank
122 behaviour in emerging and transition economies. Yan et al. (2024) study
123 how environmental protection taxes reshape green productivity among Chi-
124 nese listed firms; Barra and Falcone (2026) examine how institutional factors
125 condition environmental performance across a global sample of economies;
126 Heyert and Weill (2025) and Bach et al. (2025) use firm- and bank-level data
127 across dozens of countries to study, respectively, trust-driven financial inclu-
128 sion and credit constraints’ effect on corporate tax behaviour in Vietnamese
129 firms; Li et al. (2026) study how shadow-banking engagement reshapes trade-
130 credit financing among Chinese non-financial firms; and Horvath et al. (2025)
131 apply Bayesian model averaging to the cross-country determinants of finan-
132 cial development, a methodological precedent we build on directly in adopting

133 a Bayesian estimation strategy in Section 4. This paper sits at the intersec-
134 tion of these threads—institutions, green performance, and firm-level financ-
135 ing behaviour, studied jointly rather than pairwise—and, to our knowledge,
136 is the first among them to test whether a specific, dateable policy-adoption
137 event (SBFN membership) moderates the finance-green relationship at firm
138 level across a comparably broad cross-section of adopting economies.

139 The remainder of the paper is organized as follows. Section 2 sets out
140 the institutional background of the SBFN framework and the concepts and
141 trends in green banking policy adoption across our sample. Section 3 devel-
142 ops the paper’s three hypotheses from the state-capacity and institutional-
143 complementarity literature. Section 4 describes the primary and global sam-
144 ples, variable construction, and the empirical strategy for each. Section 5
145 presents baseline, hierarchical, and heterogeneous-effect results on the pri-
146 mary sample, followed by the independent global-sample replication. Sec-
147 tion 7 discusses the findings’ economic magnitude and their implications for
148 how green credit policy is designed and sequenced, and Section 8 concludes.

149 **2. Institutional background: green banking policy in emerging and** 150 **transition economies**

151 The Sustainable Banking and Finance Network (SBFN)—established in
152 2012 as the Sustainable Banking Network and renamed in 2019 to reflect an
153 expanded mandate covering both banking and non-bank financial regulation—
154 is the principal vehicle through which financial-sector regulators and banking
155 associations in emerging and developing economies have coordinated the de-
156 sign of national sustainable-finance policy. Hosted and technically supported

157 by the International Finance Corporation, the network now counts roughly
158 seventy member institutions (central banks, banking regulators, and banking
159 or securities associations) drawn overwhelmingly from Sub-Saharan Africa,
160 East Asia and the Pacific, Europe and Central Asia, Latin America and
161 the Caribbean, the Middle East and North Africa, and South Asia (Inter-
162 national Finance Corporation / Sustainable Banking and Finance Network,
163 2025). Notably for our sample, the network’s membership is essentially dis-
164 joint from the European Union: not one of the sixteen EU member states in
165 our WBES sample appears on any SBFN membership roster we could locate,
166 consistent with the network’s explicit mandate to serve markets outside the
167 reach of the EU’s own sustainable-finance architecture (the Sustainable Fi-
168 nance Disclosure Regulation and the EU Taxonomy, both of which post-date
169 most of our survey waves in any case).

170 Membership itself, however, is only the network’s coarsest measure of
171 policy adoption. SBFN’s own Multi-Tiered Progress Framework—the in-
172 strument used in its periodic Global Progress Reports (2018, 2019, 2021,
173 2023–2024) and country-level Progress Report addenda—scores each mem-
174 ber jurisdiction’s actual regulatory output against several policy pillars, most
175 centrally environmental-and-social risk management and, in later report vin-
176 tages, ESG disclosure integration and green-finance-flow measurement, and
177 places each jurisdiction on a progression from an initial “Commitment” or
178 “Formulating” stage, through “Developing” or “Implementation,” to “Advanc-
179 ing” and, for the most mature frameworks, “Consolidating.” The variation
180 this produces across our sample is considerable and is itself part of the paper’s
181 identifying variation, not just a robustness afterthought. Mongolia’s Bank

182 of Mongolia was a founding SBFN member in 2012 and had already reached
183 the network’s “Advancing” implementation stage by the time of the country’s
184 2019 Enterprise Survey; Morocco’s Bank Al-Maghrib, a member since 2014,
185 sat at the same stage two years later. Bangladesh Bank, also a founding 2012
186 member, had progressed by its 2022 Enterprise Survey wave into the net-
187 work’s ESG-integration pillar at an “Advancing” classification, while India’s
188 Reserve Bank/Indian Banks’ Association, a later (2016) entrant, remained
189 at a “Developing” commitment-pillar stage at its own 2022 survey. Jordan
190 and the Kyrgyz Republic, both members since 2016 and 2018 respectively,
191 had progressed only as far as an initial “Commitment” stage by their 2019
192 survey waves. A further, and for identification purposes equally important,
193 group of economies in our sample joined SBFN only after their Enterprise
194 Survey was fielded—Kazakhstan (2021), Serbia and Ukraine (2020), Tajik-
195 istan (2023), and most of the Western Balkan and South Caucasus economies
196 in our sample (2021–2024)—and are accordingly coded as non-adopters as of
197 their survey year, even though several are now members. This ex-ante cod-
198 ing is deliberate: it is the policy environment a firm actually operated under
199 at the moment its Green Economy module responses were recorded that is
200 relevant to our test, not the policy environment that jurisdiction eventually
201 adopted.

202 The economic logic behind the SBFN framework, and the reason it is a
203 plausible channel for firm-level green investment rather than a purely sym-
204 bolic commitment, rests on a straightforward supervisory mechanism: once
205 a banking regulator formally adopts environmental-and-social risk manage-
206 ment guidelines, commercial banks under its supervision face a compliance

incentive to price environmental risk into individual lending decisions and, in many of the more advanced frameworks, to report a green-lending share to the supervisor directly. Existing evidence on China’s much-studied green credit guidelines documents exactly this channel operating at bank and firm level—reduced lending to high-polluting borrowers and improved environmental performance among firms that retain access to credit (Li et al., 2023; Huang et al., 2023; Dai et al., 2025; Gao et al., 2022)—and a smaller literature on Bangladesh’s own long-running green-banking guidelines points in the same direction (Khairunnessa et al., 2021). What neither literature has established is whether this channel is detectable across a broad, heterogeneous cross-section of SBFN-style adopters at the firm level using a common survey instrument, nor whether the strength of the channel depends on the regulatory capacity available to enforce the underlying guideline—the two questions this paper turns to next.

3. Literature review and hypothesis development

3.1. Access to finance and firm-level green investment

A firm’s decision to install pollution-abatement equipment, retrofit machinery for energy efficiency, or absorb the upfront cost of an emissions-monitoring system is, first and foremost, an investment decision, and investment decisions are shaped by whether a firm can finance them. The theoretical case is close to a standard credit-constraint story: green investments of the kind captured in the WBES Green Economy module (heating and cooling upgrades, on-site renewable generation, waste-minimization retrofits, energy-management systems) typically carry a large upfront capi-

231 tal outlay against a payback stream realized only over several years, which is
 232 exactly the profile of investment most sensitive to a firm’s access to external
 233 finance rather than to retained earnings alone. Firm-level evidence bears
 234 this out directly. Ghosh and Dutta (2022), using a panel of Indian manu-
 235 facturing firms, show that credit-constrained firms are measurably less likely
 236 to adopt cleaner production practices even when they report an equivalent
 237 willingness to comply with environmental regulation. De Haas et al. (2025)
 238 document, across a large cross-country firm sample, that managerial and fi-
 239 nancial barriers operate as distinct but jointly binding constraints on firms’
 240 green transition investments, and Kalantzis et al. (2022) reach a closely re-
 241 lated conclusion using EBRD transition-economy survey data, finding that
 242 firms’ green investment is driven by a mix of financing conditions and cli-
 243 mate exposure rather than by climate motivation alone—precisely the joint
 244 finance-and-institutions framing this paper tests directly. Costa et al. (2024)
 245 and Truong (2026) find, respectively in OECD and Vietnamese data, that
 246 financing constraints blunt firms’ green investment response to environmen-
 247 tal policy pressure that would otherwise induce it, while Aristei and Gallo
 248 (2023) show that firms with better access to credit are both more likely to
 249 have adopted green management practices before a shock and better able
 250 to sustain them through one. Ullah (2025) narrows this further to unlisted
 251 SMEs specifically, showing that financial constraints bind corporate environ-
 252 mental responsibility at smaller firm scales—a firm-size channel this paper’s
 253 own causal-forest evidence (Section 5) also implicates, though, as we report
 254 there, without recovering a clean monotonic size gradient of its own. Siewers
 255 et al. (2024) extend the firm-level environmental-performance literature to

256 a global-value-chain setting, showing that a firm’s position in international
 257 production networks shapes its environmental performance net of domes-
 258 tic financing conditions, and Bambe et al. (2026) show that environmental
 259 policy stringency itself reshapes firm efficiency in developing economies, un-
 260 derscoring that the firm-level margin, not only the aggregate or bank-level
 261 one, is where much of this literature’s true test now lies. Taken together, this
 262 literature converges on a simple first-order proposition, which any test of a
 263 green-credit-policy channel must first establish before asking whether policy
 264 moderates it:

265 **H1.** Firms with access to formal bank credit (a credit line or
 266 loan from a financial institution) are more likely to have adopted
 267 green production practices than otherwise-similar firms without
 268 such access.

269 *3.2. Does green banking policy strengthen the finance-to-green-investment*
 270 *channel?*

271 H1 describes a channel, not a policy. The entire premise of the SBFN
 272 framework and its national analogues—and of the much larger literature on
 273 green credit guidelines in China specifically (Li et al., 2023, 2024; Huang
 274 et al., 2023)—is that a supervisory mandate directing banks to price envi-
 275 ronmental risk into lending, or to preferentially finance green activity, should
 276 widen the gap in green-investment behaviour between firms with and with-
 277 out bank access relative to a counterfactual world without the policy: banks
 278 in adopting jurisdictions have a compliance incentive to seek out, and price
 279 favourably, exactly the borrowers undertaking the kind of investment our

outcome variable measures, while credit-constrained firms in the same jurisdiction see no comparable improvement in their financing terms. This is a moderation hypothesis, not a simple main-effect one, and it is worth being explicit that most of the existing supporting evidence—again, overwhelmingly from China (Dai et al., 2025; Gao et al., 2022) plus a smaller literature on Bangladesh (Khairunnessa et al., 2021), Vietnam (Nguyen and Phan, 2025), and cross-country fintech-credit channels (Tran et al., 2026)—is drawn from single-country studies of a single, unusually well-resourced green-credit regime. Whether the same interaction holds across the far more heterogeneous set of jurisdictions that have adopted an SBFN-style framework, many of them at a far earlier and less resourced stage of implementation than China’s, is untested. This motivates:

H2. The positive association between bank credit access and green-practice adoption (H1) is stronger in economies that had, as of the survey year, adopted an SBFN sustainable-finance policy framework than in economies that had not.

3.3. *Institutional complementarity: does policy credibility depend on state capacity?*

A green banking guideline is only as effective as the supervisory apparatus behind it. This is the institutional-complementarity argument at the centre of the comparative-institutions, law-and-finance, and state-capacity literatures. Hall and Soskice (2001)’s varieties-of-capitalism framework is the foundational statement of the underlying logic in its most general form: formally similar policy instruments produce systematically different outcomes depend-

304 ing on how they interlock with the surrounding institutional architecture—
 305 a regulatory mandate that functions as intended within one configuration
 306 of complementary institutions may function quite differently, or not at all,
 307 transplanted into another. Applied more narrowly to financial regulation
 308 specifically, the same logic runs through the law-and-finance and state-capacity
 309 literatures: La Porta et al. (1998) and the subsequent finance-and-growth
 310 literature (Levine, 2005; Demirgüç-Kunt and Maksimovic, 1998; Beck et al.,
 311 2005, 2008) established that formally identical financial-contracting rules pro-
 312 duce very different real-economy financing outcomes depending on the quality
 313 of the legal and regulatory institutions that enforce them, while Acemoglu
 314 et al. (2001) showed the same logic operating at the level of a country’s en-
 315 tire institutional endowment. Applied to our setting, the argument is direct:
 316 a bank supervisor’s environmental-risk guideline can only reshape lending
 317 behaviour to the extent that the supervisor has the regulatory capacity—
 318 examination authority, credible sanctions, reliable disclosure enforcement—
 319 to make compliance with it more than nominal. Where that capacity is
 320 weak, an SBFN framework may exist on paper (a formal “Commitment” or
 321 “Formulating” stage classification, in the network’s own terminology) with-
 322 out meaningfully altering bank lending behaviour on the ground, precisely
 323 the state-capacity/policy-credibility distinction that motivates this special
 324 issue. This is also the logic behind the small but growing strand of work on
 325 institutions and environmental performance published in this journal specifi-
 326 cally: Barra and Falcone (2026), studying a global sample of economies, find
 327 that institutional-quality indicators condition environmental performance in
 328 ways that are not reducible to income level alone, a result our design is built

329 to test at the firm rather than the country level. We proxy regulatory ca-
330 pacity with the World Bank’s Worldwide Governance Indicators Regulatory
331 Quality estimate, matched to each economy’s survey year, and hypothesize
332 a three-way interaction:

333 **H3.** The policy-moderation effect described in H2 is itself in-
334 creasing in regulatory quality: SBFN adoption strengthens the
335 credit-to-green-investment channel most where the supervising
336 jurisdiction has the institutional capacity to make the underly-
337 ing guideline credible, and least—potentially not at all—where it
338 does not.

339 Hypotheses H2 and H3 are, by construction, statements about how a
340 country-level variable reshapes a firm-level relationship—precisely the two-
341 level structure that Section 4 argues a classical single-level fixed-effects re-
342 gression is poorly suited to recover with a country-cluster count in the low
343 forties, and that motivates the hierarchical and machine-learning estimators
344 we turn to after reporting that classical benchmark.

345 *3.4. An auxiliary hypothesis: firm-level absorptive capacity as an alternative*
346 *margin*

347 H2 and H3 are both, in the end, statements that a *country-level* char-
348 acteristic conditions the strength of the credit-to-green channel. A separate
349 strand of the financing-constraints literature suggests the conditioning vari-
350 able that matters most may instead sit at the *firm* level: Ullah (2025) shows
351 that financial constraints bind corporate environmental responsibility specif-
352 ically at smaller firm scales, consistent with a standard fixed-cost logic in

353 which the up-front capital outlay a green retrofit requires (Section 3) is pro-
354 portionately heavier, and therefore harder to absorb even once financed, for
355 a smaller firm than a larger one. We did not set out to test this as a fourth
356 ex-ante hypothesis on a par with H1–H3; it emerged from the causal forest’s
357 feature-importance decomposition (Section 5.4), which we report honestly
358 here as an auxiliary, exploratory hypothesis motivated by that same fixed-
359 cost logic rather than one we pre-specified:

360 **H4 (auxiliary, exploratory).** Whatever heterogeneity exists
361 in the strength of the credit-to-green-adoption channel is more
362 attributable to firm size than to either country-level institutional
363 moderator (SBFN status or regulatory quality).

364 We flag its exploratory status explicitly because it changes how its evi-
365 dence should be read: Section 5.4’s support for H4 is genuine and, we think,
366 informative, but it is evidence discovered during analysis rather than con-
367 firmed against a pre-registered prediction, and the same section reports hon-
368 estly that it does not yet extend to a clean, monotonic size gradient. H4
369 also connects this paper’s evidence to the just-transition and SME-financing
370 literature’s broader concern with who is left behind when a policy instru-
371 ment operates on an extensive banking margin rather than on the firm-scale
372 margin that determines whether a financed firm can absorb a green invest-
373 ment’s fixed costs at all—a distributional dimension of green-credit-policy
374 design this paper’s data can only gesture toward, not test directly, but which
375 we think is a natural extension for future work using firm-level data with
376 a continuous size measure and a larger, more granular firm-size distribution
377 than the Enterprise Surveys’ own three-category strata provide.

378 4. Data and methodology

379 4.1. Data sources and sample construction

380 The paper’s firm-level data come from the World Bank Enterprise Sur-
381 veys (WBES), a standardized, stratified-random-sample survey of formal-
382 sector, non-agricultural, non-extractive private firms with at least five em-
383 ployees, conducted by the World Bank’s Enterprise Analysis Unit. Our pri-
384 mary analysis sample is the World Bank’s own harmonized cross-country
385 file combining the 41 Europe and Central Asia (ECA) and Middle East and
386 North Africa (MENA) economies surveyed under a single, identical question-
387 naire between 2018 and 2020, the survey wave in which the Bank’s Green
388 Economy module—a battery of questions on emissions monitoring, energy-
389 consumption targets, and the adoption of specific climate-friendly production
390 measures—was fielded in full to every respondent. This yields 28,042 firm-
391 level observations across 41 economies (Table 2). We supplement this primary
392 sample with 15,556 additional firm observations from six further economies
393 (Bangladesh 2022, India 2022, Indonesia 2023, Peru 2023, Philippines 2023,
394 Timor-Leste 2021) in which a shortened, partially randomized version of the
395 Green Economy module was fielded; because the exact item wording and ran-
396 domized sub-module assignment differ across these six economies, we do not
397 pool them into the main harmonized outcome variable, and instead use them
398 as an out-of-region external-validity check on a narrower, honestly-matched
399 pair of indicators (CO2-emissions monitoring, available in five of the six;
400 waste-minimization adoption, available in five of the six, with different spe-
401 cific economies covered by each), reported separately in Section 5.

402 The primary sample, while broad by the standards of the existing green-

403 credit-policy literature, is not the full universe of WBES economies carrying
 404 *any* Green Economy content, and we build a second, independent sample
 405 to exploit the rest of it. The World Bank also maintains a standardized
 406 cross-country database (2006–2026) that harmonizes a common set of core
 407 variables across every Formal Sector Enterprise Survey wave it has ever
 408 fielded, 292,314 firm-level observations across 471 country-years. Within
 409 it, 160 country-years fielded between 2021 and 2026 — reaching, notably,
 410 into high-income economies entirely outside the SBFN’s remit such as Aus-
 411 tralia, Canada, Germany, and Japan — carry a harmonized minimal Green
 412 Economy item set: climate-damage exposure and, more relevantly for our
 413 outcome construct, CO2-emissions monitoring, alongside the same standard
 414 access-to-finance module (credit line/loan, overdraft, perceived finance ob-
 415 stacle) used in the primary sample. We supplement this with Bangladesh
 416 2022 and Indonesia 2023 — two economies whose Green Economy responses
 417 used an alternative randomized-module variable naming convention and so
 418 are absent from the standardized database’s harmonized columns, but which
 419 we can recover from our own extraction of the underlying country files —
 420 for a combined global sample of 162 economies and 100,115 firms (89,797
 421 with complete data on all covariates). Firm size, sector, and productivity
 422 controls for this sample are drawn from the Bank’s companion Productivity
 423 Database, merged by firm identifier. We refer to this as the *global sample*,
 424 and to CO2-emissions monitoring as its outcome variable throughout; it is
 425 the paper’s vehicle for testing whether the primary sample’s central find-
 426 ing replicates independently, at far greater scale, and under an identification
 427 strategy — country fixed effects — that the primary sample’s single-wave

428 design cannot support (Section 4).

429 Country-level data are drawn from two sources for both samples. For the
430 primary sample’s 47 economy-years, SBFN policy-adoption status, join year,
431 and progression-framework stage as of each economy’s survey year are com-
432 piled from the Sustainable Banking and Finance Network’s Global Progress
433 Reports (2018, 2019, 2021, 2023–2024), country-level Progress Report ad-
434 denda, and the network’s live country-profile data portal, cross-checked against
435 IFC press releases announcing individual member accessions; economies that
436 joined SBFN only after their WBES survey year are coded as non-adopters
437 as of that year (Section 2). For the global sample’s remaining 157 economy-
438 years, we draw membership status and join year directly from the SBFN data
439 portal’s full published member roster (67 members as of the portal’s most
440 recent update), applying the identical ex-ante rule: an economy is coded as
441 an adopter only if its join year precedes its survey year. Six economy-years
442 in the global sample (Central African Republic, Cameroon, Chad, Republic
443 of Congo, Equatorial Guinea, and Gabon, all 2023–2024) are recorded on
444 the roster under a regional entry, “Central African States,” which we inter-
445 pret as referring to the CEMAC zone’s common banking supervisor; four
446 small-island economy-years (Antigua and Barbuda, Grenada, St. Lucia, St.
447 Vincent and the Grenadines, all 2025) are similarly recorded under “East-
448 ern Caribbean States,” which we interpret as the Eastern Caribbean Central
449 Bank’s member states. Both interpretations are flagged explicitly here as
450 inferences rather than explicit per-country listings. Regulatory capacity is
451 measured with the World Bank’s Worldwide Governance Indicators (WGI)
452 Regulatory Quality estimate, matched to each economy’s exact survey year

453 (or, where not yet published, the most recent prior year available) via the
454 World Bank DataBank API; WGI does not cover Taiwan, which is accord-
455 ingly dropped from global-sample regressions.

456 A natural alternative to the cross-sectional design used throughout this
457 paper would exploit SBFN’s own staggered adoption timing—join years in
458 our sample span 2012–2024 (Section 2)—in a staggered difference-in-differences
459 or event-study design around each economy’s own adoption date, the identi-
460 fication strategy this literature (and this special issue specifically) most often
461 asks for. We do not attempt this, and it is worth being explicit about why
462 rather than leaving the gap implicit. The WBES Green Economy module
463 is not a repeated panel instrument: it has been fielded to a given country,
464 in most cases, exactly once, as a special one-off module attached to a single
465 survey wave, and even the small subset of economies in our sample surveyed
466 more than once by the WBES core instrument were not re-surveyed with a
467 comparable Green Economy module in a second wave; the six supplementary
468 economies in our own extension sample (Section 5.5) illustrate the problem di-
469 rectly, since even they differ from one another, and from the primary sample,
470 in exact item wording and randomized sub-module assignment, precluding a
471 consistently defined outcome measured twice for the same country before and
472 after its SBFN join date. A within-country before/after comparison around
473 each economy’s own adoption date is accordingly not implementable with a
474 consistent outcome variable in the currently published WBES microdata, for
475 this specific outcome construct. This is a genuine data-availability constraint
476 on the identification strategy available to us, not a design choice; we flag it
477 here as a first-class limitation of the paper’s identification, and return to its

478 consequences in Section 8.

479 4.2. Variable construction

480 **Outcome.** Our primary dependent variable, *green practice adoption*, is
481 a binary indicator equal to one if a firm reports having adopted at least one
482 of seven Green Economy module practices over the preceding three years:
483 more climate-friendly on-site energy generation, an energy-management sys-
484 tem, waste-minimization or recycling measures, air-pollution control mea-
485 sures, other pollution-control measures, any energy-efficiency measure, or
486 the use of on-site renewable energy. We also construct a continuous *green*
487 *adoption index*, the mean of the same seven binary items, for robustness.

488 **Treatment.** *Bank credit access* is a binary indicator equal to one if the
489 firm reports holding a line of credit or loan from a formal financial institution
490 at the time of the survey.

491 **Country-level moderators.** *SBFN member* is a binary indicator equal
492 to one if the economy had adopted a national SBFN sustainable-finance
493 policy framework as of the survey year. *Regulatory quality* is the WGI Reg-
494 ulatory Quality estimate for the survey year (continuous, standardized to
495 mean zero, unit standard deviation within the estimation sample for the
496 hierarchical and machine-learning specifications).

497 **Controls.** Firm size (small/medium/large, per the Enterprise Surveys’
498 own sampling strata), broad sector (manufacturing, services, construction),
499 log sales, an exporter dummy (any direct or indirect export share), and a
500 foreign-ownership dummy (any private foreign ownership share).

501 Table 1 lists every variable used in the analysis together with its exact
502 WBES question code and source, in the interest of full transparency about

503 variable construction given the multi-source nature of the merge.

504 4.3. Empirical strategy

505 We estimate the relationship between bank credit access and green prac-
 506 tice adoption, and its moderation by SBFN status and regulatory quality,
 507 in five stages of increasing methodological sophistication, each addressing a
 508 specific limitation of the one before it.

509 **Stage 1 — classical benchmark.** We first estimate a pooled logistic
 510 regression,

$$\Pr(\text{Green}_{i,c} = 1) = \Lambda\left(\beta_1 \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + \delta_{s(i)}\right) \quad (1)$$

511 for firm i in country c , where $\mathbf{X}_{i,c}$ is the firm-control vector and $\delta_{s(i)}$ is a
 512 sector fixed effect, with standard errors clustered by country. We deliberately
 513 do not include country fixed effects in this specification: because SBFN_c is
 514 constant within country in a single cross-sectional wave, a country fixed effect
 515 would absorb it—and its interaction with $\text{Credit}_{i,c}$ —completely, making β_2
 516 and β_3 unidentifiable by construction. This is not a modelling choice we
 517 are free to make differently; it is the reason a purely classical fixed-effects
 518 approach cannot answer H2/H3 as posed, and we report this specification,
 519 extended with a triple interaction for $\text{Regulatory quality}_c$, precisely to make
 520 that limitation visible.

521 **Stage 2 — Bayesian hierarchical (multilevel) model.** We re-
 522 estimate the same relationship allowing country to enter as a random rather

523 than fixed effect, with a random slope on Credit:

$$\text{Green}_{i,c} \sim \text{Bernoulli}(p_{i,c}) \quad (2)$$

$$\begin{aligned} \text{logit}(p_{i,c}) = & (\beta_1 + u_{1,c}) \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) \\ & + \beta_4 (\text{Credit}_{i,c} \times \text{RegQuality}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + (\alpha + u_{0,c}) \end{aligned} \quad (3)$$

$$\begin{pmatrix} u_{0,c} \\ u_{1,c} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad (4)$$

524 estimated via Hamiltonian Monte Carlo (the No-U-Turn Sampler, NUTS)
 525 in PyMC. This specification is the paper’s primary vehicle for testing H2
 526 and H3: because country enters through a variance component rather than
 527 a full set of dummies, the country-level moderators SBFN_c and RegQuality_c
 528 remain identified, while the random slope $u_{1,c}$ still absorbs unobserved coun-
 529 try heterogeneity in the credit-green relationship and, together with partial
 530 pooling across countries, gives more defensible inference than cluster-robust
 531 standard errors computed over only 41 clusters (Cameron and Miller, 2015).

532 **Stage 3 — causal forest / double machine learning.** To avoid
 533 imposing a linear interaction functional form on how the treatment effect
 534 of credit access varies across the institutional landscape, we additionally es-
 535 timate a Causal Forest with Double Machine Learning (Wager and Athey,
 536 2018; Chernozhukov et al., 2018; Athey et al., 2019), using gradient-boosted
 537 nuisance models for the outcome and treatment propensity, and firm size, ex-
 538 port status, SBFN status, and regulatory quality as effect-modifying features.
 539 This lets us recover conditional average treatment effects (CATEs)—the esti-
 540 mated effect of credit access on green adoption, separately for SBFN members
 541 versus non-members, and separately by regulatory-quality tercile—without

542 imposing that the moderation take the single-coefficient interaction form as-
 543 sumed in Stages 1–2. We use “causal” here in the qualified sense standard
 544 in this literature: the forest’s identifying assumption is unconfoundedness
 545 conditional on the observed covariate set (firm size, export status, SBFN
 546 status, regulatory quality), not an exogenous source of variation in credit
 547 access itself. This is exactly the same assumption underlying the logit and
 548 hierarchical specifications; Stage 3 relaxes the functional form imposed on
 549 that assumption from linear to fully flexible, but it does not relax the as-
 550 sumption, and the CATEs reported below should be read accordingly as
 551 doubly-robust conditional associations rather than causal effects identified
 552 off a natural experiment.

553 **Stage 4 — small-sample supplementary check.** We additionally
 554 re-estimate a reduced-form version of the credit-green relationship using a
 555 waste-minimization-adoption outcome, available in four economies (India
 556 2022, Indonesia 2023, Peru 2023, Timor-Leste 2021) whose Green Economy
 557 modules asked this item under directly comparable wording. This outcome
 558 is not part of the standardized cross-country database underlying Stage 5
 559 below, so it offers genuinely independent, if small-sample, supplementary
 560 evidence.

561 **Stage 5 — global-sample replication with country fixed effects.**
 562 Finally, and centrally, we re-estimate the full specification on the 162-economy
 563 global sample described in Section 4, using CO2-emissions monitoring as the
 564 outcome. Because this sample spans far more country clusters than the
 565 primary sample, we can now afford a country-fixed-effects specification: δ_c
 566 absorbs SBFN_c and RegQuality_c ’s main effects (collinear with the fixed effect

by construction, since both are constant within country-year) while leaving their firm-level interactions with $\text{Credit}_{i,c}$ fully identified, since credit access itself varies within country. This is a materially sharper test of H2/H3 than Stage 1's clustered-SE approach, on an entirely independent sample and outcome from Stages 1–3; we report it alongside a no-fixed-effects specification for direct comparability with the primary sample's Table 3.

Table 1: Variable definitions and sources.

Variable	Definition	WBES code	Source
green_adoption_binary	=1 if firm adopted ≥ 1 of 7 Green Economy practices in last 3 years	BMGc23b, c23d, c23e, c23f, c23j, c25, e5	WBES Green Economy Module (2018–2020)
fin_has_credit_line	=1 if firm has a line of credit or loan	K8	WBES Access to Finance Module
sbfn_member	=1 if SBFN policy adopted as of survey year	n/a	SBFN Global Progress Reports; data.sbfnetwork.org
wgi_regulatory_quality	WGI Regulatory Quality estimate, survey year	n/a	World Bank DataBank API (GOV_WGI_RQ_EST)
size_cat	Firm size stratum (Small/Medium/Large)	A6A	WBES Sampling Frame
sector_broad	Broad sector (Manufacturing/Services/Construction)	A4AX	WBES Sampling Frame
log_sales	Log of total annual sales	D2	WBES General Information Module
foreign_owned_dummy	=1 if any foreign ownership share $> 0\%$	B2B	WBES Ownership Module
exporter_dummy	=1 if any export sales share $> 0\%$	D3B, D3C	WBES Sales Module

5. Results

5.1. Descriptive patterns

Table 2: Sample composition, by economy.

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Albania 2019	377	2019	0	0.09	0.609	0.407
Armenia 2020	546	2020	0	0.21	0.684	0.482
Azerbaijan 2019	225	2019	0	-0.10	0.418	0.180
Belarus 2018	600	2018	0	-0.79	0.662	0.425
Bosnia and Herze- govina 2019	362	2019	0	-0.38	0.581	0.565
Bulgaria 2019	772	2019	0	0.46	0.604	0.467
Croatia 2019	404	2019	0	0.38	0.708	0.448
Cyprus 2019	240	2019	0	1.04	0.746	0.711
Czechia 2019	502	2019	0	1.12	0.725	0.526
Egypt 2020	3,075	2020	1	-0.64	0.303	0.065
Estonia 2019	360	2019	0	1.54	0.738	0.412
Georgia 2019	581	2019	1	0.88	0.441	0.501
Greece 2018	600	2018	0	0.29	0.825	0.499
Hungary 2019	805	2019	0	0.46	0.673	0.537
Italy 2019	760	2019	0	0.70	0.506	0.230
Jordan 2019	601	2019	1	0.16	0.471	0.248
Kazakhstan 2019	1,446	2019	0	0.11	0.519	0.257
Kosovo 2019	271	2019	0	-0.38	0.698	0.425
Kyrgyz Republic 2019	360	2019	1	-0.42	0.606	0.299
Latvia 2019	359	2019	0	1.00	0.824	0.386

Economy	N firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Lebanon 2019	532	2019	0	-0.56	0.452	0.365
Lithuania 2019	358	2019	0	0.98	0.678	0.342
Malta 2019	242	2019	0	0.83	0.901	0.479
Moldova 2019	360	2019	0	-0.11	0.619	0.328
Mongolia 2019	360	2019	1	-0.07	0.625	0.492
Montenegro 2019	150	2019	0	0.29	0.387	0.576
Morocco 2019	1,096	2019	1	0.03	0.478	0.251
North Macedonia 2019	360	2019	0	0.19	0.572	0.508
Poland 2019	1,369	2019	0	0.83	0.569	0.403
Portugal 2019	1,062	2019	0	1.00	0.812	0.600
Romania 2019	814	2019	0	0.36	0.438	0.474
Russia 2019	1,323	2019	0	-0.35	0.503	0.310
Serbia 2019	361	2019	0	0.06	0.654	0.585
Slovak Republic 2019	429	2019	0	0.73	0.782	0.386
Slovenia 2019	409	2019	0	0.88	0.793	0.674
Tajikistan 2019	352	2019	0	-0.80	0.523	0.194
Tunisia 2020	615	2020	1	-0.11	0.391	0.413
Türkiye 2019	1,663	2019	1	-0.02	0.346	0.345
Ukraine 2019	1,337	2019	0	-0.29	0.715	0.244
Uzbekistan 2019	1,239	2019	0	-0.92	0.751	0.240

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
West Bank and Gaza 2019	365	2019	0	-0.15	0.428	0.178

Note: Reg. qual. is the WGI Regulatory Quality estimate for the survey year. Green rate is the unconditional share of firms reporting adoption of at least one Green Economy practice. Credit rate is the share of firms reporting a bank credit line or loan.

575 Table 2 reports the full 41-economy sample composition; firm counts range
576 from 150 (Montenegro) to 3,075 (Egypt), and the raw, unconditional green-
577 practice-adoption rate ranges more than three-fold across economies, from
578 30.3% (Egypt) to 90.1% (Malta). Figure 1 maps this variation alongside
579 SBFN policy status: 8 of the 41 primary-sample economies had adopted
580 an SBFN framework as of their survey year (Georgia, Egypt, Jordan, Kyr-
581 gyz Republic, Mongolia, Morocco, Tunisia, and Türkiye)—13 of the full 47-
582 economy sample once the five (of six) SBFN-member small-sample supple-
583 mentary economies are counted—and every European Union member state
584 in the sample is, consistent with SBFN’s mandate, a non-adopter. At the
585 firm level, bank credit access correlates positively with the green-adoption
586 index ($r = 0.18$) but is essentially uncorrelated with a firm’s own perceived
587 finance obstacle ($r = -0.02$)—a first hint, well before any regression, that re-
588 alized access to credit rather than perceived financing difficulty is what tracks
589 green behaviour, and the two are evidently not interchangeable measures of
590 the same underlying constraint.

591 Figure 2 previews the paper’s central finding graphically before a sin-
592 gle regression is estimated: plotting each economy’s green-adoption rate

593 against its regulatory-quality score and fitting separate lines for SBFN mem-
 594 bers and non-members produces two lines of very similar, modestly positive
 595 slope, separated by a roughly 15–20 percentage-point vertical gap—SBFN-
 596 member economies show systematically *lower* raw green-adoption rates at
 597 every level of regulatory quality, not a steeper relationship between regu-
 598 latory quality and adoption. That combination—a level difference with no
 599 slope difference—is exactly the signature the regression evidence below con-
 600 firms formally.

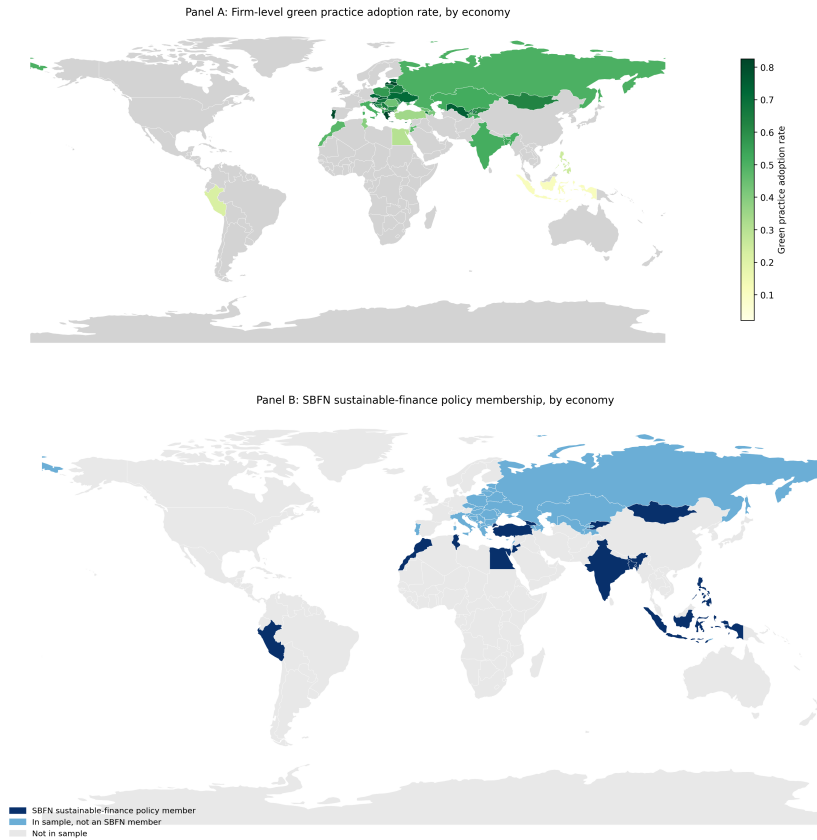


Figure 1: Green practice adoption and SBFN policy membership, by economy.

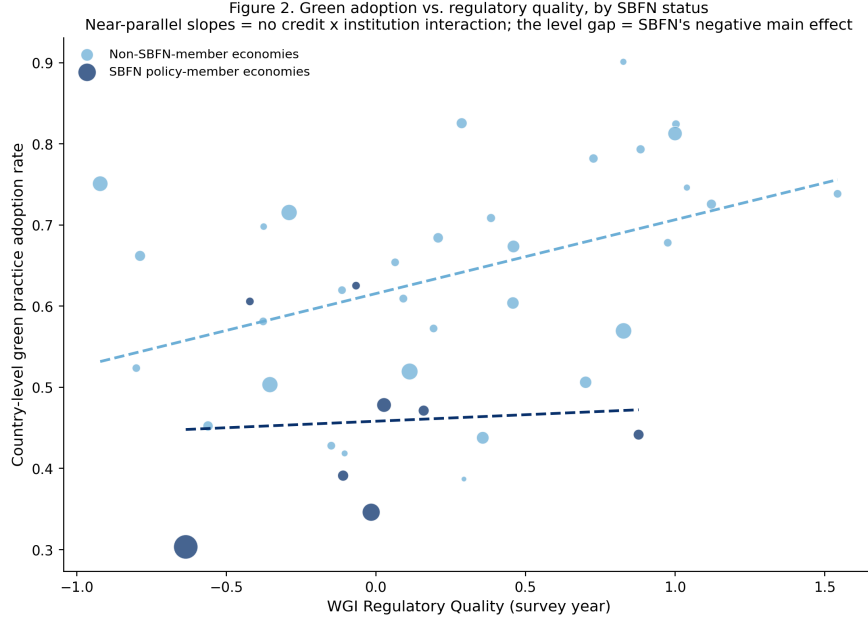


Figure 2: Green adoption vs. regulatory quality, by SBFN status.

5.2. Baseline classical benchmark

Table 3 reports the pooled logistic baseline. Column M1 confirms H1 on its own terms: firms with a bank credit line are significantly more likely to have adopted a green practice ($b = 0.571$, $p < 0.001$; average marginal effect = 12.95 percentage points, 95% CI [7.7, 18.2])—holding a formal credit line is associated with roughly the same increase in the probability of green adoption as moving from a small to a large firm, and a somewhat larger increase than being an exporter. Column M2 adds SBFN status and its interaction with credit access: the SBFN main effect is large, negative, and precisely estimated ($b = -0.991$, $p < 0.001$), while the credit $\times$ SBFN interaction is small and statistically indistinguishable from zero ($b = 0.049$, $p = 0.773$). Column M3's fully saturated triple interaction with regulatory quality tells the same story

at greater length: every interaction term involving credit access is small and insignificant, while regulatory quality’s uninteracted relationship with adoption is positive, if only marginally significant in this particular fully-saturated specification ($b = 0.278$, $p = 0.110$).

As flagged in Section 4, this specification cannot include country fixed effects without mechanically absorbing the very SBFN and regulatory-quality terms the hypotheses concern, so what Table 3 shows is a genuine test of H2/H3, not a placeholder—and that test returns no interaction. The question the remaining two stages are built to answer is whether that null result reflects a real absence of policy moderation, or an artefact of forcing a two-level question through a single-level model with country-clustered standard errors computed over only 41 clusters, a setting in which such standard errors are known to be unreliable (Cameron and Miller, 2015).

5.3. Bayesian hierarchical model — primary test of H2/H3

The hierarchical specification—country-varying random slopes on credit access, cross-level interactions with SBFN status and standardized regulatory quality, four chains of 1,000 post-warmup draws each, all $\hat{R} \leq 1.01$ and ess_{bulk} between 797 and 2,270—is built specifically to give H2 and H3 a fair hearing that the absorbed classical specification could not. It does not change the conclusion. Table 4 reports the full posterior summary; three results stand out.

First, the firm-level credit effect (H1) survives unchanged in substance: posterior mean 0.33 (89% equal-tailed interval $[0.18, 0.49]$), comfortably excluding zero. Second, both country-level institutional variables retain clean, precisely estimated *main* effects on the baseline adoption rate—SBFN mem-

Table 3: Baseline classical logit regressions (dependent variable: green practice adoption, binary). Country-clustered standard errors in parentheses.

	M1	M2	M3
Credit access	0.571*** (0.123)	0.459*** (0.090)	0.373*** (0.085)
SBFN member		-0.991*** (0.178)	-0.844*** (0.209)
Credit $\times$ SBFN		0.049 (0.168)	0.019 (0.140)
Regulatory quality			0.278 (0.174)
Credit $\times$ Reg. quality			0.144 (0.142)
SBFN $\times$ Reg. quality			0.053 (0.261)
Credit $\times$ SBFN $\times$ Reg. quality			-0.104 (0.272)
Log sales	0.061 (0.039)	0.041 (0.034)	0.068** (0.032)
Foreign owned	0.219 (0.143)	0.244* (0.139)	0.220 (0.143)
Exporter	0.537*** (0.092)	0.471*** (0.074)	0.412*** (0.079)
Sector, size controls	Yes	Yes	Yes
N	23,914	23,914	23,914
Pseudo R^2	0.058	0.089	0.093

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

bership net-negative (-0.57, 89% ETI [-1.00, -0.09]), regulatory quality net-positive (0.34, 89% ETI [0.16, 0.53])—reproducing, with proper hierarchical uncertainty quantification rather than an absorbed or clustered approximation, exactly the level-shift pattern Figure 2 shows visually. Third, and centrally for H2 and H3, neither cross-level interaction is distinguishable from zero: credit $\times$ SBFN, 0.072 (89% ETI [-0.26, 0.40]); credit $\times$ regulatory quality (standardized), 0.004 (89% ETI [-0.13, 0.14])—a near-exact zero on the latter, not merely an imprecise one.

We read this as an honest and, we think, informative finding rather than a disappointing one: giving the moderation hypothesis a hierarchical model built for precisely this two-level question still returns no interaction, while sharpening (relative to the pooled baseline) the confidence with which we can state that SBFN status and regulatory quality shift the *level* of green adoption directly.

Table 4: Bayesian hierarchical model, posterior summary (four chains, 1,000 post-warmup draws each).

Parameter	Mean	SD	89% ETI low	89% ETI high	$\hat{R}$
Intercept	-1.93	0.246	-2.30	-1.50	1.00
Credit access	0.33	0.097	0.18	0.49	1.00
SBFN member	-0.57	0.300	-1.00	-0.09	1.01
Credit $\times$ SBFN	0.072	0.211	-0.26	0.40	1.00
Reg. quality (std.)	0.344	0.116	0.16	0.53	1.00
Credit $\times$ Reg. quality	0.004	0.087	-0.13	0.14	1.00

652 5.4. Causal forest / heterogeneous treatment effects

653 Table 5 and Figure 5 report the Causal Forest DML estimates, which im-
654 pose no linear-interaction functional form and let the treatment effect vary
655 flexibly across firm and country covariates. The overall average treatment
656 effect of credit access is 0.073 (95% CI [-0.040, 0.186])—directionally consis-
657 tent with, though smaller in magnitude and less precisely estimated than,
658 the logit average marginal effect of 0.130 in Section 5.2, reflecting the more
659 conservative variance properties of a doubly-robust, cross-fitted nonparamet-
660 ric estimator relative to a parametric logit on the same data. Critically,
661 the conditional average treatment effects triangulate with Section 5.3 rather
662 than contradicting it: the estimated effect is 0.072 for firms in non-SBFN
663 economies versus 0.076 for firms in SBFN economies, and 0.075 / 0.070 /
664 0.075 across the low/middle/high regulatory-quality terciles respectively—
665 three different institutional cuts, three essentially flat effects, all visibly over-
666 lapping in Figure 5. Two entirely independent estimators, one parametric
667 and hierarchical, one nonparametric and forest-based, agree on the absence
668 of institutional moderation.

669 The forest’s feature-importance decomposition adds a genuinely new re-
670 sult rather than only confirming the null, and speaks directly to the auxiliary
671 H4 introduced in Section 3.4: of the four candidate effect-modifiers, firm size
672 (log sales) accounts for 75.6% of the model’s explained heterogeneity, versus
673 19.9% for regulatory quality, 2.6% for SBFN status, and 1.9% for exporter
674 status. Whatever heterogeneity exists in how strongly credit access pre-
675 dicts green adoption is, on this evidence, primarily a firm-level rather than a
676 country-institutional phenomenon, supporting H4. Collapsing this into the

Enterprise Surveys’ own coarse size strata, however, does not reveal a clean monotonic gradient: the estimated effect is 0.080 for small firms, 0.075 for medium firms, and 0.057 for large firms (Table 6; all with wide, overlapping confidence intervals spanning zero), which if anything runs opposite to a simple “larger firms better absorb a green investment’s fixed costs” story rather than confirming it. Read alongside the 75.6% importance share for the continuous size measure, we take this as evidence that the heterogeneity the forest is detecting is more granular or non-monotonic across the firm-size distribution than a three-category collapse can reveal, rather than evidence for any particular direction of a size gradient—a qualification we report explicitly rather than paper over with a tidier-sounding but unsupported claim. This redirects, rather than closes, the paper’s institutional-moderation question, and we return to it in Section 7.

5.5. *Small-sample supplementary check: waste-minimization adoption*

Before turning to the global-sample replication that is this paper’s second major test (Section 5.7), Table 7 reports a smaller supplementary check using waste-minimization adoption — an outcome not captured in the standardized cross-country database underlying the global sample, and so a genuinely independent (if small-sample) data point — available in four economies whose Green Economy modules asked this item under directly comparable wording: India 2022, Indonesia 2023, Peru 2023, and Timor-Leste 2021. Three of these four are themselves SBFN members at various progression stages, so this sample cannot re-test the SBFN-adoption contrast; what it can test is whether the credit-to-green channel documented in Sections 5.2–5.4 is a peculiarity of the ECA-MENA sample or generalizes elsewhere. It generalizes:

Table 5: Causal forest DML: average treatment effect, CATEs, and feature importances.

Estimate	Value	95% CI low	95% CI high
Overall ATE	0.0733	-0.0396	0.1862
CATE SBFN non-member	0.0723	-0.0300	0.1746
CATE SBFN member	0.0758	-0.0591	0.2107
CATE reg. quality tercile: low	0.0753	-0.0568	0.2075
CATE reg. quality tercile: mid	0.0697	-0.0355	0.1749
CATE reg. quality tercile: high	0.0749	-0.0203	0.1701

Feature	Importance
SBFN member	0.0257
Regulatory quality	0.1993
Log sales	0.7562
Exporter dummy	0.0189

Table 6: Causal forest CATEs by firm-size category.

Size category	CATE	95% CI low	95% CI high
Small	0.0797	-0.0409	0.2004
Medium	0.0747	-0.0344	0.1838
Large	0.0569	-0.0428	0.1565

702 pooled across the four economies, having an overdraft facility is associated
703 with significantly higher odds of waste-minimization adoption ($b = 0.422$,
704 $p < 0.001$, $N = 9,038$, 4 countries). The direction of the core finance-green
705 relationship replicates on a different continent, a different half-decade, and
706 a narrower outcome definition than the one anchoring Sections 5.2–5.4 — a
707 preview, on four economies, of the much larger replication in Section 5.7.

Table 7: Small-sample supplementary check: descriptive rates.

Economy	N	Waste-min. rate	Overdraft rate	SBFN member
India 2022	9,376	0.515	0.621	1
Indonesia 2023	2,955	0.336	0.144	1
Peru 2023	987	0.806	0.549	1
Timor-Leste 2021	238	0.061	0.089	0

708 5.6. Additional robustness

709 Table 9 subjects the primary-sample finding to four further checks, all
710 retaining sector fixed effects, firm controls, and country-clustered standard
711 errors (no country fixed effects, for the reason given in Section 4). First,
712 replacing the binary outcome with the continuous seven-item green-adoption
713 index and estimating by OLS reproduces the same pattern to the decimal:
714 credit access positive and significant ($b = 0.053$, $p < 0.001$), SBFN main
715 effect negative and significant ($b = -0.092$, $p < 0.001$), interaction indistin-
716 guishable from zero ($b = 0.006$, $p = 0.739$).

717 Second, replacing bank credit access with two alternative finance mea-
718 sures produces one genuine point of nuance worth reporting rather than
719 smoothing over. Using overdraft-facility access instead of a credit line/loan,

720 the main overdraft effect remains positive and significant ($b = 0.274$, $p =$
 721 0.011), the SBFN main effect remains negative and significant ($b = -0.905$,
 722 $p < 0.001$)—but the overdraft $\times$ SBFN interaction is now negative and sig-
 723 nificant ($b = -0.346$, $p = 0.031$), the one interaction term in the entire paper
 724 that clears a conventional significance threshold, and it runs opposite to the
 725 direction H2 predicts. Using the reverse-coded perceived finance-obstacle
 726 measure in place of realized access, by contrast, both the main effect and
 727 its SBFN interaction are precisely zero ($b = -0.010$, $p = 0.827$; interac-
 728 tion $b = -0.012$, $p = 0.849$)—consistent with the descriptive correlation
 729 noted in Section 5.1. We read the overdraft result as suggestive rather than
 730 conclusive given it is one significant coefficient among more than a dozen
 731 interaction tests across the paper, but it is at least directionally consistent
 732 with a substantive story worth flagging for future work: if SBFN-guided
 733 lending is channelled preferentially through the term loans and credit lines
 734 that finance long-horizon capital expenditure, rather than through short-term
 735 overdraft facilities, we might expect exactly this pattern—a null-to-positive
 736 interaction on credit lines and a negative one on overdrafts—though our
 737 data cannot directly test whether SBFN-guided credit is disproportionately
 738 term-structured.

739 Third, splitting the sample into manufacturing ($N = 13,340$) and services
 740 ($N = 9,538$) subsamples shows the credit-access main effect, the SBFN main
 741 effect, and the null credit $\times$ SBFN interaction all replicate within each sector
 742 separately (manufacturing: credit $b = 0.480$, $p < 0.001$, interaction $b =$
 743 -0.040 , $p = 0.826$; services: credit $b = 0.448$, $p < 0.001$, interaction $b =$
 744 0.195 , $p = 0.426$)—the paper’s central null is not an artefact of pooling two

745 structurally different production technologies.

746 Because the overdraft interaction (the one coefficient in the paper that
747 clears a conventional significance threshold) is one of many credit $\times$ institutional-
748 moderator tests reported across the frequentist specifications in this paper,
749 we additionally apply a Benjamini–Hochberg false-discovery-rate correction
750 across the full family of 13 such coefficients reported under a classical (non-
751 Bayesian, non-forest) estimator in Tables 3, 9, and 10 (Table 8; full calcu-
752 lation reported in the replication package). The overdraft interaction’s raw
753 $p = 0.031$ becomes $p = 0.403$ after correction, and no coefficient in the
754 family survives at a false discovery rate of 0.10 or even 0.40. We read this
755 as reinforcing, with a formal multiple-testing accounting rather than only a
756 qualitative caveat, the suggestive-not-conclusive framing we already give this
757 result in Section 7: an isolated nominally-significant interaction among more
758 than a dozen tested is exactly what a pure false-discovery process would be
759 expected to produce roughly one time in twenty, and this correction does not
760 let us reject that account.

761 5.7. *Global-sample replication with country fixed effects*

762 The preceding five subsections establish the paper’s central finding on the
763 primary sample. We now subject that finding to what we consider the most
764 demanding test available to us: complete re-estimation on an independently
765 assembled, 162-economy global sample (Section 4), using a different outcome
766 variable (CO2-emissions monitoring rather than the seven-item composite),
767 different survey years (2021–2026 rather than 2018–2020), and, for the first
768 time in the paper, a country-fixed-effects specification that this much larger
769 set of clusters can support. Figure 3 maps the combined reach of the two

Table 8: Multiple-testing correction: Benjamini–Hochberg false discovery rate applied to every credit $\times$ institutional-moderator interaction coefficient reported under a frequentist estimator (Tables 3, 9, 10). Table 3 model M3 entries marked $\dagger$ are approximate p -values recovered from the reported coefficient and standard error via a normal approximation ($z = \text{coef}/\text{se}$), since the text reports only the coefficient and standard error for these terms.

Source	Term	Raw p	BH-adjusted p
Table 9 (b1) Overdraft	Overdraft $\times$ SBFN	0.031	0.403
Table 10 M2 (country FE)	Credit $\times$ SBFN	0.205	0.799
Table 10 M1 (no country FE)	Credit $\times$ Reg. quality †	0.301	0.799
Table 3 M3 †	Credit $\times$ Reg. quality	0.310	0.799
Table 10 M2 (country FE)	Credit $\times$ Reg. quality	0.346	0.799
Table 9 (c2) Services	Credit $\times$ SBFN	0.426	0.799
Table 10 M1 (no country FE)	Credit $\times$ SBFN	0.430	0.799
Table 3 M3 †	Credit $\times$ SBFN $\times$ Reg. quality	0.702	0.892
Table 9 (a) Continuous index, OLS	Credit $\times$ SBFN	0.739	0.892
Table 3 M2	Credit $\times$ SBFN	0.773	0.892
Table 9 (c1) Manufacturing	Credit $\times$ SBFN	0.826	0.892
Table 9 (b2) Finance obstacle (reverse-coded)	Obstacle $\times$ SBFN	0.849	0.892
Table 3 M3 †	Credit $\times$ SBFN	0.892	0.892

Table 9: Additional robustness checks.

Specification	Finance coef. (p)	SBFN coef. (p)	Interaction coef. (p)
(a) Continuous index, OLS	0.053 (0.000)	-0.092 (0.000)	0.006 (0.739)
(b1) Overdraft	0.274 (0.011)	-0.905 (0.000)	-0.346 (0.031)
(b2) Finance obstacle (reverse-coded)	-0.010 (0.827)	-1.011 (0.000)	-0.012 (0.849)
(c1) Manufacturing subsample	0.480 (0.000)	-1.047 (0.000)	-0.040 (0.826)
(c2) Services subsample	0.448 (0.000)	-0.914 (0.001)	0.195 (0.426)

770 samples together: 166 distinct economies, spanning every region and, through
771 the global sample specifically, extending for the first time in this literature
772 into high-income economies with no SBFN involvement at all.

773 Table 10 reports two specifications. Column M1, without country fixed
774 effects, is the direct global-sample analogue of Table 3’s primary-sample base-
775 line: credit access remains positive and significant ($b = 0.198$, $p = 0.039$,
776 $N = 89,797$, 159 countries), while the SBFN main effect ($b = -0.171$,
777 $p = 0.353$), the credit $\times$ SBFN interaction ($b = 0.278$, $p = 0.430$), and
778 every regulatory-quality interaction are statistically indistinguishable from
779 zero. Column M2 imposes country fixed effects, absorbing the SBFN and
780 regulatory-quality main effects by construction (they are collinear with δ_c ,
781 since both are constant within a given country-year) while leaving their firm-
782 level interactions with credit access fully identified, because credit access
783 itself varies within country. This is a sharper test than anything the primary
784 sample’s single 2018–2020 wave can support, and it returns the same answer
785 with even more precision: credit access is strongly significant ($b = 0.249$,
786 $p < 0.001$), while the credit $\times$ SBFN interaction ($b = 0.506$, $p = 0.205$) and
787 the credit $\times$ regulatory-quality interaction ($b = -0.044$, $p = 0.346$) remain
788 null.

789 We regard this as the paper’s most important robustness result. It is not
790 a re-analysis of the same data under a different label: it is a different outcome
791 construct, a different set of economies (only a handful of which overlap with
792 the primary sample), a different half-decade, and an econometric specification
793 — country fixed effects — that is only available at all because this second
794 sample has enough clusters to support it. That the qualitative conclusion

795 is identical across both samples and across the full set of four estimators
796 reported in this paper (classical logit, Bayesian hierarchical, causal forest, and
797 now country-fixed-effects logit) is, we think, considerably stronger evidence
798 for the paper’s central claim than either sample could offer alone.

Figure 3. Combined sample coverage: 166 economies (41 primary rich-module + 162 global minimal-indicator, net of overlap)

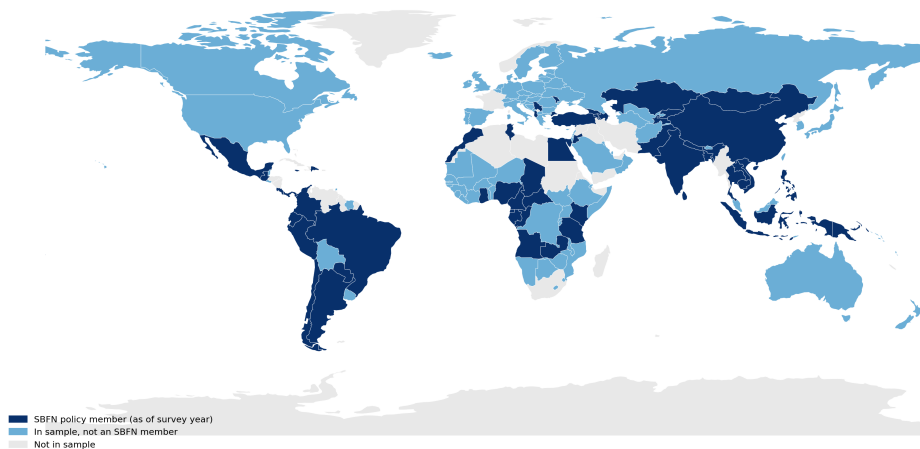


Figure 3: Combined sample coverage: 166 economies (41 primary + 162 global, net of overlap), shaded by SBFN status.

799 5.8. Results summary across all specifications

800 Sections 5.2–5.7 report ten tables spanning four estimators and two inde-
801 pendent samples. Figure 4 consolidates every coefficient bearing on H1–H3
802 into a single view. Panel A shows the credit-access effect (H1): every point
803 estimate is positive and every confidence interval excludes zero, in both sam-
804 ples and all four estimators. Panels B and C show the credit $\times$ SBFN and
805 credit $\times$ regulatory-quality interactions (H2, H3): every single confidence in-
806 terval, across both samples and every estimator, straddles zero. Figure 5
807 shows the same non-result from a completely different angle—the causal

Table 10: Global-sample regressions (dependent variable: CO2-emissions monitoring, binary). Country-clustered standard errors in parentheses; M2's SBFN and regulatory-quality main effects are absorbed by country fixed effects and are not separately estimable.

	M1: no country FE	M2: country FE
Credit access	0.198** (0.096)	0.249*** (0.057)
SBFN member	-0.171 (0.184)	— (absorbed)
Credit $\times$ SBFN	0.278 (0.352)	0.506 (0.399)
Regulatory quality (std.)	0.155 (0.096)	— (absorbed)
Credit $\times$ Reg. quality	0.092 (0.089)	-0.044 (0.047)
Log sales	0.127*** (0.027)	0.288*** (0.028)
Foreign owned	0.621*** (0.083)	0.528*** (0.046)
Exporter	0.456*** (0.094)	0.335*** (0.044)
Country fixed effects	No	Yes
N	89,797	89,797
Countries	159	159

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Taiwan dropped (no WGI coverage).

808 forest's conditional average treatment effects broken out by SBFN status,
809 regulatory-quality tercile, and firm-size category all overlap both each other
810 and the overall average treatment effect. No subgroup split, by any variable
811 examined in this paper, recovers a distinguishable effect.

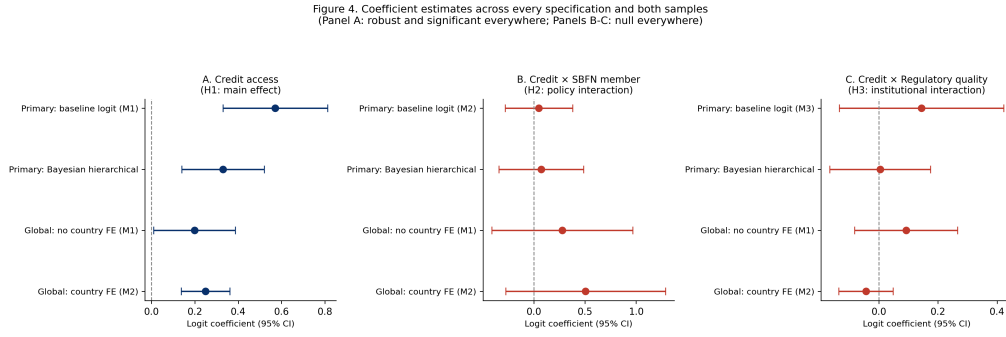


Figure 4: Coefficient estimates across every specification and both samples.

812 6. Supplementary evidence: a country-level staggered event-study 813 on macro environmental outcomes

814 Sections 5.2–5.8 establish this paper's central finding using firm-level
815 cross-sectional data, for the data-availability reasons given in Section 4: the
816 WBES Green Economy module cannot support a staggered difference-in-
817 differences or event-study design around each economy's own SBFN adop-
818 tion date, because it has not been fielded as a repeated panel instrument.
819 That constraint applies to the firm-level finance-to-green-investment channel
820 specifically; it does not apply to a different, complementary question that
821 the same SBFN adoption-timing variation can answer directly: does a coun-
822 try's aggregate environmental trajectory shift around its own SBFN adoption

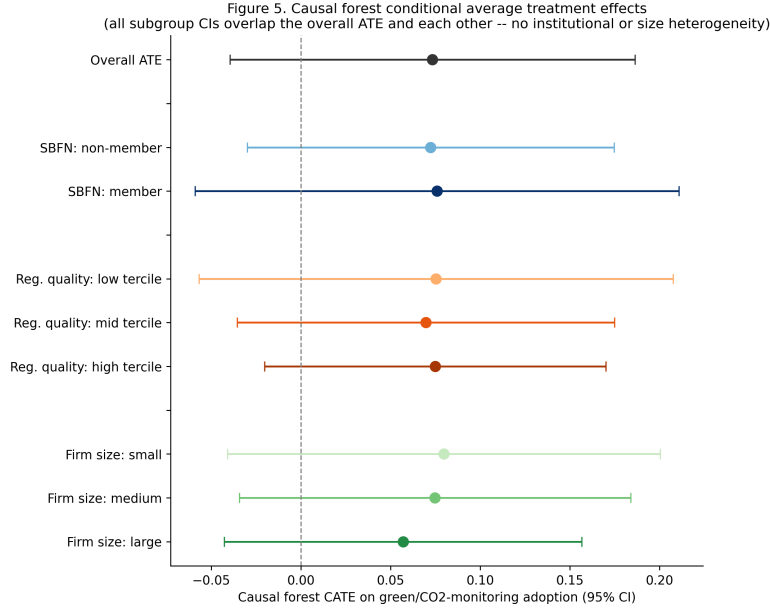


Figure 5: Causal forest conditional average treatment effects, by subgroup.

date? This section builds and reports that test, using an entirely independent data source and unit of analysis, precisely because it is the identification strategy this literature – and this special issue in particular – asks for, and because a genuine answer, however it comes out, is more useful than leaving the question unaddressed.

6.1. Data and design

We assemble a country-year panel spanning every World Bank member economy with a non-aggregate ISO3 code (217 economies), 2000–2024. Treatment timing is each economy’s SBFN join year, drawn from the same SBFN data portal membership roster used to construct the global sample (Section 4) and cross-checked against the global-sample SBFN coding used in Section 5.7 (zero disagreements across 61 overlapping economies); economies

835 that have never joined SBFN, as of the roster’s most recent update, serve as
 836 the never-treated comparison group. This yields 68 ever-treated economies
 837 (join years 2012–2024, the same real staggered-adoption variation described
 838 in Section 2) and 149 never-treated economies. Outcome data come from
 839 the World Bank’s World Development Indicators, retrieved via the World
 840 Bank DataBank API: renewable energy consumption (% of total final en-
 841 ergy consumption, `EG.FEC.RNEW.ZS`) and CO2 emissions per capita (AR5
 842 GWP-consistent series, `EN.GHG.CO2.PC.CE.AR5`), both available for the large
 843 majority of economies from 2000 through at least 2021–2024. We addition-
 844 ally draw on GDP per capita (constant 2015 US\$, `NY.GDP.PCAP.KD`) as a
 845 covariate, to net out the obvious confound that SBFN-adopting economies
 846 are disproportionately still-industrializing, faster-growing economies whose
 847 emissions profiles are shaped by their development stage independently of
 848 any banking-sector policy.

849 We estimate the effect of SBFN adoption on each outcome using the
 850 Callaway and Sant’Anna (2021) staggered-adoption difference-in-differences
 851 estimator (Callaway and Sant’Anna, 2021), which recovers group-time av-
 852 erage treatment effects for each adoption cohort separately and aggregates
 853 them into an overall average treatment effect on the treated (ATT) and
 854 an event-study profile, using only never-treated economies as the compar-
 855 ison group. We report this estimator, rather than a standard two-way-
 856 fixed-effects (TWFE) regression, because TWFE is now well known to pro-
 857 duce badly biased estimates under staggered treatment timing when already-
 858 treated units are implicitly used as controls for later-treated units (Goodman-
 859 Bacon, 2021); we report a naive TWFE specification alongside the Callaway-

860 Sant'Anna estimates specifically to illustrate the size of that bias in our own
 861 data; rather than as a candidate preferred estimate. For each outcome we
 862 report both an unconditional specification and a doubly-robust specification
 863 conditioning on log GDP per capita.

864 6.2. Results

865 Table 11 reports the overall ATT for both outcomes under three speci-
 866 fications; Figure 6 plots the full event-study profile (doubly-robust, GDP-per-
 867 capita-controlled specification) for both outcomes.

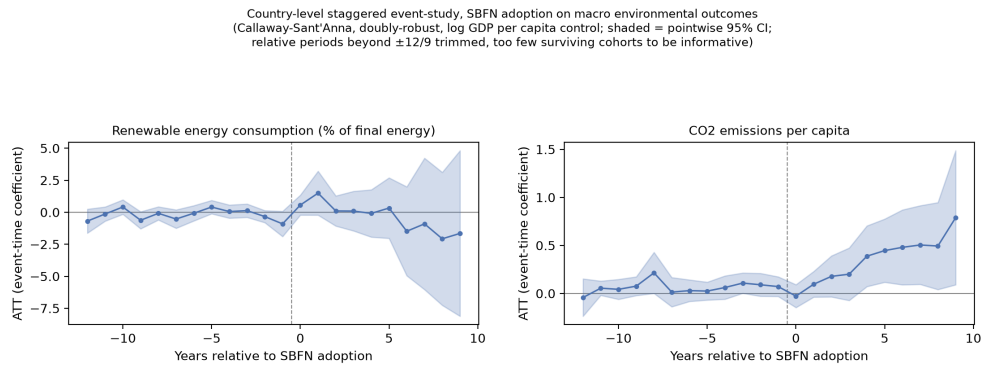


Figure 6: Country-level staggered event-study, doubly-robust specification with log GDP per capita control. Shaded bands are pointwise 95% confidence intervals; relative periods beyond $\pm 12/9$ are trimmed for having too few surviving cohorts to be informative.

868 Two results, read together, do not support treating SBFN adoption as a
 869 country-level lever that visibly improves aggregate environmental outcomes –
 870 which is itself informative, since this is the identification strategy best suited
 871 to answer that specific question directly. First, renewable energy share shows
 872 no effect distinguishable from zero in either specification, and the doubly-
 873 robust event-study profile (Figure 6, left panel) is flat and precisely estimated

Table 11: Country-level staggered event-study: overall average treatment effect of SBFN adoption on macro environmental outcomes. Callaway-Sant’Anna estimator, never-treated comparison group, 999 bootstrap iterations; naive TWFE reported for comparison only. Full country-year panel, 2000–2024.

Outcome	Specification	ATT	SE	95% CI
Renewable energy %	Callaway-Sant’Anna, no co- variates	-1.192	0.802	[-2.763, 0.380]
Renewable energy %	Callaway-Sant’Anna, doubly- robust, log GDP p.c.	0.674	0.916	[-1.122, 2.470]
Renewable energy %	Naive TWFE (post dummy, country+year FE)	-4.672***	1.127	[-6.882, -2.463]
CO2 per capita	Callaway-Sant’Anna, no co- variates	0.631***	0.179	[0.280, 0.982]
CO2 per capita	Callaway-Sant’Anna, doubly- robust, log GDP p.c.	0.349**	0.156	[0.044, 0.655]
CO2 per capita	Naive TWFE (post dummy, country+year FE)	0.901***	0.204	[0.500, 1.302]

*** 95% CI excludes zero; ** 95% CI excludes zero, narrower margin. Underlying data and code in the replication package.

874 on both sides of the adoption date: a clean null, not an underpowered one.

875 Second, CO2 emissions per capita shows a positive (i.e., the “wrong-
876 signed” direction if SBFN adoption reduced emissions) and statistically sig-
877 nificant association with adoption in every specification, but three features
878 of the result argue against reading it causally. It shrinks by roughly 45%
879 (0.631 to 0.349) once we condition on log GDP per capita, indicating that
880 a substantial share of the raw association is attributable to exactly the
881 development-stage confound we set out to net out. The doubly-robust event-
882 study profile (Figure 6, right panel) shows a small, mostly flat pre-period
883 with only two marginally significant leads, followed by a post-period effect
884 that grows smoothly and monotonically from near zero at adoption to its
885 largest magnitude nine to twelve years out – a continuing-divergence pattern
886 more consistent with SBFN-adopting economies already sitting on a different,
887 faster-rising emissions trajectory than non-adopters (which a single covariate
888 cannot fully absorb) than with a discrete policy-caused jump at the adop-
889 tion date. And the naive TWFE estimate (0.901) is itself more than double
890 the doubly-robust Callaway-Sant’Anna estimate (0.349), the same direction
891 of bias documented in the staggered-DID literature (Goodman-Bacon, 2021)
892 and a useful internal illustration of why we lead with the more robust esti-
893 mator throughout this section.

894 We are deliberately not spinning this into either a “green banking pol-
895 icy backfires” or a “green banking policy works” claim. The honest reading
896 is that a macro-level design – built using exactly the identification strategy
897 (staggered adoption timing, a proper Callaway-Sant’Anna estimator, never-
898 treated controls) this special issue’s own methodological expectations call

899 for – also does not produce clean evidence that SBFN adoption changes a
900 country’s aggregate environmental trajectory for the better, and the one sta-
901 tistically significant result we do find is more parsimoniously explained by
902 which economies self-select into SBFN membership than by what the policy
903 itself does once adopted. This is, we think, a genuinely different and com-
904plementary piece of evidence to Sections 5.2–5.7’s firm-level results, not a
905restatement of them: different data source, different unit of analysis, differ-
906ent outcome constructs, and, for the first time in this paper, an identification
907strategy that exploits exogenous variation in adoption *timing* rather than re-
908lying on cross-sectional comparison across adopters and non-adopters. That
909it converges on the same substantive conclusion – no clean evidence of a ben-
910eficial institutional effect – from a design built specifically to meet a bar the
911paper’s firm-level evidence cannot meet, strengthens rather than substitutes
912for the paper’s central finding.

913 7. Discussion

914 Put in plain economic terms, the paper’s central finding is this: a firm
915 holding a bank credit line is associated with an adoption probability roughly
916 13 percentage points higher than an otherwise-similar firm without one—
917 an association net of the observed firm and country controls, not a causally
918 identified effect, for the reasons given in Limitation 1 below—comparable in
919 magnitude to the gap between a small firm and a large one—and that gap is
920 essentially the same size whether the firm sits in an SBFN sustainable-finance
921 policy adopter or a non-adopter, and essentially the same size whether the
922 country’s regulatory quality sits in the bottom, middle, or top third of the

923 sample (Figure 4). Firms in SBFN-adopting economies are, on average, some
924 15–20 percentage points *less* likely to have adopted a green practice at all—a
925 level gap, not a slope gap.

926 This convergence matters most because of how differently the two sam-
927 ples could have failed to agree. The primary and global samples share al-
928 most no economies in common, were fielded in different half-decades, measure
929 green practice adoption with entirely different survey items, and reach insti-
930 tutional settings—high-income, non-SBFN economies such as Australia and
931 Germany—that the primary sample cannot touch at all. A finding that held
932 only in the 41-country ECA-MENA sample could plausibly be a regional or
933 period-specific artefact; a finding that survives unchanged when re-estimated
934 on 162 largely non-overlapping economies under a sharper identification strat-
935 egy is considerably harder to explain away as such.

936 Section 6’s country-level staggered event-study adds a third, indepen-
937 dently identified data point to this convergence, and arguably the most de-
938 manding one: a completely different data source (World Bank World Devel-
939 opment Indicators rather than firm surveys), a completely different unit of
940 analysis (country-year rather than firm), and, for the first time in the pa-
941 per, an identification strategy that exploits exogenous variation in adoption
942 *timing* specifically—the design this special issue’s own methodological guid-
943 ance asks for. That it too finds no clean evidence of a beneficial institutional
944 effect (a precise null on renewable-energy share, and a positive CO2 associa-
945 tion that shrinks substantially under a development-stage control and whose
946 dynamics look more like pre-existing divergence than a policy-caused shift)
947 is, we think, the single hardest-to-dismiss piece of evidence in the paper that

948 the null on institutional moderation is a real feature of how SBFN adoption
949 relates to environmental outcomes, not an artefact of the firm-level data’s
950 own identification limits.

951 This is worth dwelling on precisely because it complicates rather than
952 confirms the implicit premise of much of the existing green-credit-policy lit-
953 erature, which is disproportionately built on evidence from China’s unusu-
954 ally mature, unusually well-resourced green credit guidelines (Li et al., 2023;
955 Huang et al., 2023; Dai et al., 2025). Our sample includes economies at ev-
956 ery stage of the SBFN’s own progression framework—from Jordan and the
957 Kyrgyz Republic’s initial “Commitment” stage to Mongolia’s “Advancing” im-
958 plementation a full seven years after founding membership—and across that
959 entire range, we find no evidence that formal policy adoption changes how a
960 firm’s credit access translates into green investment behaviour. Section 5.4’s
961 most novel result gives one plausible redirection of the question, though not
962 a tidy answer to it: the heterogeneity that *does* exist in the credit-green
963 relationship is overwhelmingly a firm-size phenomenon (75.6% of explained
964 variance) rather than a country-institutional one (regulatory quality 19.9%,
965 SBFN status a mere 2.6%)—yet, as Section 5.4 reports honestly, collaps-
966 ing this into the Surveys’ own coarse Small/Medium/Large strata does not
967 recover a clean gradient in either direction, so we cannot yet say whether
968 it is small or large firms for whom the credit-green channel runs stronger,
969 only that firm scale—in some more granular or non-monotonic form than a
970 three-category split reveals—is where the forest locates most of the effect het-
971 erogeneity a country-level policy variable does not. What this does support is
972 a reorientation of where policy attention belongs: if the operative constraint

sits at the level of which individual firms can absorb a green investment’s fixed costs once financed, rather than at the level of whether a supervisory guideline nudges banks to lend more broadly, then a country-level supervisory mandate operating on the extensive margin of bank lending decisions may be aimed at the wrong margin regardless of its own institutional credibility—a question future work with a finer-grained firm-size or capacity measure is better placed to resolve than the coarse categories available here.

It is worth being precise about what this null does and does not establish. “No interaction detected” and “no institutional complementarity exists” are not the same claim, and we do not want the paper’s repeated confirmation of the former to be read as having settled the latter. At least three distinct readings remain open. First, our preferred reading throughout the paper: the complementarity genuinely does not operate through this channel. Second, a linear interaction term may simply be the wrong functional form to detect a genuinely threshold-shaped complementarity—SBFN’s own Multi-Tiered Progress Framework (Section 2) treats policy adoption as a graded process from an initial “Commitment” stage through “Consolidating,” and our binary membership coding (Limitation 2 below) could easily average over a complementarity that only activates once a jurisdiction clears a particular progression stage, in a way four estimators built around a single membership dummy cannot recover. Third, the Worldwide Governance Indicators’ broad, economy-wide regulatory-quality measure may simply not proxy the specific banking-supervisory capacity a green-lending guideline actually depends on (Limitation 5); a null on this proxy is evidence about the proxy’s channel, not necessarily about supervisory capacity in general. We think the first

998 reading is the best-supported of the three given the evidence in Section 5,
999 but we hold it as the best current account rather than as a claim the paper
1000 has definitively ruled the other two out.

1001 A further alternative worth engaging directly, since it is the more theo-
1002 retically interesting institutional-mechanism story a system-change-oriented
1003 reading of our results would want addressed: SBFN guidelines could, in prin-
1004 ciple, be working exactly as intended on the *extensive* margin of who obtains
1005 credit at all, rather than on the *intensive* margin our interaction terms test.
1006 If a green-banking guideline induces banks in adopting jurisdictions to extend
1007 credit more broadly to environmentally promising firms that would otherwise
1008 have gone unfinanced, that effect shows up as a shift in *which* firms hold a
1009 credit line, not as a different adoption gap between financed and unfinanced
1010 firms conditional on already holding one—exactly the quantity our credit $\times$
1011 SBFN interaction estimates. Testing this directly would require modelling
1012 credit access itself, rather than green adoption conditional on credit access,
1013 as a function of SBFN status and firm environmental characteristics; our
1014 design, built around the demand-side finance-to-green channel, does not do
1015 this, and we flag it explicitly as an unaddressed alternative mechanism rather
1016 than one our null result rules out.

1017 We should also be candid about a limit of our outcome measure itself.
1018 Every green-practice-adoption item in this paper is firm-reported, and we
1019 cannot rule out that self-reported adoption rates differ systematically from
1020 banks’ own compliance reporting to their SBFN supervisor in ways our
1021 data cannot observe—a general concern about compliance-reporting-versus-
1022 practice gaps that applies to any policy evaluated through the regulated

1023 party’s own attestations, and one a firm-level survey instrument, however
1024 carefully harmonized, cannot fully resolve on its own.

1025 To our knowledge, this paper is also the first to test the SBFN frame-
1026 work’s own theory of change against firm-level, rather than bank-level or
1027 policy-stage, outcomes: the network’s periodic Global Progress Reports (Sec-
1028 tion 2) measure and publicize each member jurisdiction’s regulatory-output
1029 stage, but not whether that output changes what happens on the ground
1030 for the firms the policy ultimately targets. Read this way, our null result is
1031 directly relevant to the SBFN’s and the IFC’s own self-evaluation practice, in-
1032 dependent of what it implies for the academic institutional-complementarity
1033 literature: a framework can progress steadily through Commitment, Develop-
1034 ing, and Advancing stages by every measure the network itself tracks, while
1035 the firm-level channel that is its ultimate point remains, on this evidence,
1036 unmoved.

1037 The one interaction that does clear a conventional significance threshold—
1038 a *negative* credit $\times$ SBFN effect when finance access is measured through
1039 overdraft facilities rather than term credit lines (Section 5)—is, we think,
1040 more consistent with this redirection than with a simple failure of green
1041 banking policy. Overdrafts finance working capital, not the kind of multi-year
1042 capital expenditure our outcome variable measures; if SBFN-guided lending
1043 is disproportionately channelled through the term-loan instruments better
1044 suited to financing a solar installation or an energy-efficiency retrofit, exactly
1045 the pattern we find—a null-to-positive effect on credit lines, a negative one
1046 on overdrafts—is what a supervisor successfully steering credit toward long-
1047 horizon green investment, rather than short-horizon working capital, would

1048 produce. We flag this as suggestive rather than established, since it is a
1049 single significant coefficient among the many interaction tests reported across
1050 Tables 3–9—and, as Section 5 reports formally (Table 8), it does not survive a
1051 Benjamini–Hochberg correction for multiple comparisons at any conventional
1052 false discovery rate—and our data cannot directly verify the underlying loan-
1053 tenor mechanism.

1054 Read against this special issue’s organizing themes of state capacity and
1055 policy credibility, and holding to the same best-current-account framing as
1056 above rather than treating it as settled, our results sit closer to a state-
1057 capacity-is-necessary-but-firm-level-frictions-bind-independently story than to
1058 a straightforward institutional-complementarity one in which stronger regu-
1059 latory quality unlocks a latent policy effect that would otherwise lie dor-
1060 mant. Regulatory quality matters—it raises the general baseline propensity
1061 to adopt green practices regardless of a firm’s own financing arrangement—
1062 but it does not appear to be the missing ingredient that would let a green
1063 banking guideline reshape the credit-to-green channel specifically. This sug-
1064 gests, tentatively, that the binding constraint on green credit policy’s effec-
1065 tiveness in our sample is less about whether the supervising jurisdiction has
1066 the state capacity to make the guideline credible, and more about whether
1067 firms below a certain scale can absorb a green investment’s fixed costs at all
1068 once credit is available—a question about firm-level absorptive capacity that
1069 sits one level down from the state-capacity question the SBFN framework
1070 itself is designed to address.

8. Conclusion

This paper set out to test whether the wave of green banking policy adoption coordinated through the Sustainable Banking and Finance Network changes the relationship between firm-level access to finance and green investment behaviour, and whether any such moderation depends on the regulatory capacity available to make the underlying policy credible. Using harmonized World Bank Enterprise Survey data spanning 41 economies’ full 2018–2020 Green Economy Module rollout, a four-country waste-minimization supplementary check, and an independently assembled 162-economy global sample spanning 2021–2026 and reaching for the first time into high-income, non-SBFN economies, we apply a deliberately escalating sequence of estimators—a classical benchmark chosen to expose its own limitation, a Bayesian hierarchical model built to give the moderation hypothesis a fair test, a causal forest imposing no functional-form assumption at all, and, on the global sample specifically, a country-fixed-effects specification the primary sample’s single wave cannot support—and find four consistent results. First, access to bank credit is robustly associated with a firm’s green practice adoption, a relationship that survives every specification, subsample, and alternative-outcome check in the paper and replicates independently in both the small waste-minimization sample and the full 162-economy global sample. Second, neither SBFN policy adoption nor regulatory quality moderates that relationship in any of the four estimators, on either sample, though both shift the general level of green adoption directly and substantially in the primary sample. Third, where genuine effect heterogeneity does exist, it is concentrated at the level of firm size rather than country institutions. Fourth, this

entire pattern replicates on a sample sharing almost no economies with the primary one, fielded years later, measured with a different outcome item, and identified under a materially sharper econometric strategy. Fifth, and most importantly for the paper’s overall credibility, an entirely independent country-level staggered event-study (Section 6), built specifically to supply the adoption-timing identification strategy the firm-level data cannot, likewise finds no clean evidence that SBFN adoption improves aggregate environmental outcomes—the single strongest piece of evidence in the paper that the null on institutional moderation is a real feature of how this policy relates to outcomes rather than an artefact of any one sample, data source, level of analysis, or estimator.

For policy, the implication is not that green banking guidelines are ineffective, but that our evidence does not support treating them as an effective lever specifically for widening the gap in green investment between financed and unfinanced firms—the margin most green-credit-policy literature implicitly assumes they operate on. If our tentative loan-tenor interpretation of the overdraft-interaction finding is correct, policymakers designing or refining SBFN-style frameworks may find more traction focusing supervisory attention on the term-lending instruments that actually finance green capital expenditure, and on complementary measures—technical assistance, matching grants, or credit guarantees—that address smaller firms’ absorptive-capacity constraints directly, rather than assuming that a bank-level supervisory mandate alone will reach firms regardless of scale.

We close with seven limitations that bound how far these conclusions travel. First, both samples are single cross-sections per country (Section 4

1121 discusses why a genuine panel on this outcome, and the staggered-adoption
 1122 design it would support, does not exist in the published WBES data): they
 1123 cannot rule out reverse causality (firms already predisposed to green invest-
 1124 ment may find it easier to obtain credit) or unobserved, time-invariant coun-
 1125 try characteristics correlated with both SBFN adoption and the baseline
 1126 propensity to adopt green practices, and our identification rests on firm-level
 1127 variation in credit access interacting with a plausibly firm-exogenous country-
 1128 level policy indicator, not a natural experiment; the global sample's country
 1129 fixed effects sharpen this only for the interaction terms, not for the credit
 1130 main effect itself. Second, SBFN membership is a binary proxy for what
 1131 the network's own multi-tiered progression framework treats as a genuinely
 1132 graded process; a natural extension would model policy stage—Commitment
 1133 through Consolidating—as an ordinal or continuous treatment rather than
 1134 collapsing it to membership status, which our binary coding necessarily does,
 1135 and which for the global sample's regional entries (the CEMAC and East-
 1136 ern Caribbean groupings, Section 4) additionally rests on an inference about
 1137 which specific economies a regional listing covers rather than an explicit
 1138 per-country statement. Third, the four-economy waste-minimization check
 1139 cannot re-test the SBFN contrast itself, since three of its four economies are
 1140 SBFN members, and combines non-identical outcome items across countries
 1141 by construction of the WBES's own shortened, randomized module design
 1142 in later survey waves—a genuine data-availability constraint rather than a
 1143 methodological choice. Fourth, the global sample's CO2-monitoring outcome
 1144 is a single item rather than the primary sample's seven-item composite, and is
 1145 a narrower construct than "green practice adoption" in the sense the primary

1146 sample measures it; that the same qualitative pattern holds on this narrower
1147 measure is reassuring but does not establish that it would hold on a richer
1148 one, were a richer one available at global scale. Fifth, the Worldwide Gover-
1149 nance Indicators’ regulatory-quality measure is a broad, economy-wide gover-
1150 nance indicator rather than a banking-supervision-specific capacity measure;
1151 a supervisor-specific index, were one available consistently across this coun-
1152 try set, would sharpen the institutional-complementarity test considerably.
1153 Sixth, our design tests only the intensive margin of the credit-to-green chan-
1154 nel (the adoption gap between financed and unfinanced firms); it cannot rule
1155 out that SBFN policy operates instead on the extensive margin of which
1156 firms obtain credit in the first place, nor can our firm-reported outcome
1157 measure distinguish actual practice adoption from a compliance-reporting
1158 gap between what firms report and what banks report to their supervisor
1159 (Section 7). Seventh, Section 6’s country-level event-study is an ecological-
1160 level test: it speaks to whether SBFN adoption shifts a country’s aggregate
1161 environmental trajectory, not to the firm-level financing mechanism the rest
1162 of the paper investigates, and the two should not be conflated. Its one
1163 significant result (CO2 per capita) is netted against a single covariate, log
1164 GDP per capita; we cannot rule out that a richer set of development-stage
1165 controls (energy-mix transition, industrial composition, population growth)
1166 would shrink the estimated association further still, or that some genuine, if
1167 small and confounded, relationship remains. We leave each of these to future
1168 work.

1169 **Declaration of generative AI and AI-assisted technologies in the**
1170 **manuscript preparation process**

1171 During the preparation of this work the author(s) used Claude (An-
1172 thropic) to assist with data extraction and cleaning from World Bank Enter-
1173 prise Surveys microdata (including identification and extraction of the stan-
1174 dardized cross-country database underlying the global sample), construction
1175 of the analysis datasets and merge with country-level SBFN and Worldwide
1176 Governance Indicators data, execution of the statistical and machine-learning
1177 analyses reported in Section 5, construction of the country-year World Devel-
1178 opment Indicators panel and the Callaway-Sant’Anna staggered event-study
1179 reported in Section 6, generation of Figures 1–3 and Figure 6, and drafting of
1180 portions of the manuscript text. After using this tool, the author(s) reviewed
1181 and edited the content as needed and take full responsibility for the content
1182 of the published article.

1183 **Appendix A. SBFN policy status and regulatory quality, full sam-**
1184 **ple**

1185 Table A.12 reports, for all 47 economy-years in the paper’s primary and
1186 small-sample supplementary check, the SBFN membership status coded for
1187 the analysis, the join year where applicable, the SBFN Multi-Tiered Progress
1188 Framework stage in force nearest the survey year, and the exact year of the
1189 Worldwide Governance Indicators observation used. Full per-country source
1190 citations (the specific SBFN Global Progress Report, country addendum, or
1191 accession announcement consulted for every determination) are available in
1192 the online supplementary material. The analogous 162-economy table for

1193 the global sample (Section 4) is summarized by region in Appendix C and
1194 provided in full in the online supplementary material and the replication
1195 package.

Table A.12: SBFN policy status and regulatory quality, full 47-
economy sample.

Economy	SBFN member	Join year	Stage at survey	WGI year
Albania 2019	0	—	None	2019
Armenia 2020	0	—	None	2020
Azerbaijan 2019	0	—	None	2019
Bangladesh 2022	1	2012	Advancing (ESG Integra- tion)	2022
Belarus 2018	0	—	None	2018
Bosnia and Herzegovina 2019	0	—	None	2019
Bulgaria 2019	0	—	None	2019
Croatia 2019	0	—	None	2019
Cyprus 2019	0	—	None	2019
Czechia 2019	0	—	None	2019
Egypt 2020	1	2016	Formulating (Prepara- tion)	2020
Estonia 2019	0	—	None	2019
Georgia 2019	1	2017	Developing (Implementa- tion)	2019
Greece 2018	0	—	None	2018
Hungary 2019	0	—	None	2019

Economy	SBFN member	Join year	Stage at survey		WGI year
India 2022	1	2016	Developing	(Commitment)	2022
Indonesia 2023	1	2012	Consolidating (Maturing)		2023
Italy 2019	0	—	None		2019
Jordan 2019	1	2016	Commitment	(Preparation)	2019
Kazakhstan 2019	0	—	None		2019
Kosovo 2019	0	—	None		2019
Kyrgyz Republic 2019	1	2018	Commitment	(Preparation)	2019
Latvia 2019	0	—	None		2019
Lebanon 2019	0	—	None		2019
Lithuania 2019	0	—	None		2019
Malta 2019	0	—	None		2019
Moldova 2019	0	—	None		2019
Mongolia 2019	1	2012	Advancing	(Implementation)	2019
Montenegro 2019	0	—	None		2019
Morocco 2019	1	2014	Advancing	(Implementation)	2019
North Macedonia 2019	0	—	None		2019
Peru 2023	1	2013	Advancing	(ESG Integration)	2023
Philippines 2023	1	2013	Advancing	(ESG Integration)	2023
Poland 2019	0	—	None		2019

Economy	SBFN member	Join year	Stage at survey	WGI year
Portugal 2019	0	—	None	2019
Romania 2019	0	—	None	2019
Russia 2019	0	—	None	2019
Serbia 2019	0	—	None	2019
Slovak Republic 2019	0	—	None	2019
Slovenia 2019	0	—	None	2019
Tajikistan 2019	0	—	None	2019
Timor-Leste 2021	0	—	None	2021
Tunisia 2020	1	2019	Unverified	2020
Türkiye 2019	1	2015	Developing (Implementation)	2019
Ukraine 2019	0	—	None	2019
Uzbekistan 2019	0	—	None	2019
West Bank and Gaza 2019	0	—	None	2019

1196 Appendix B. Bayesian model convergence diagnostics

1197 Convergence diagnostics for every focal parameter of the hierarchical
1198 model reported in Table 4 are included directly in that table ($\hat{R}$ column); the
1199 full posterior summary also includes ess_{bulk} ranging from 797 (SBFN member)
1200 to 2,270 (credit $\times$ regulatory quality), comfortably above the conventional
1201 minimum threshold of 400 effective samples per parameter for stable poste-
1202 rior summaries at four chains. No divergent transitions were flagged during

1203 sampling (NUTS via the nutpie sampler, four chains of 1,000 post-warmup
1204 draws each after 1,000 tuning iterations).

1205 **Appendix C. Global-sample SBFN status, regional summary**

1206 Of the 162 global-sample economy-years, 61 are coded as SBFN members
1207 as of their survey year and 101 as non-members. Table C.13 reports the
1208 regional distribution of the 61 members, drawn from the SBFN data por-
1209 tal’s regional classification; five of the 61 (Peru, Philippines, Timor-Leste,
1210 Bangladesh, and Indonesia) carry the deeper per-country sourcing already
1211 reported in Appendix A, having been determined during the primary sam-
1212 ple’s original research pass rather than from the portal roster alone.

Table C.13: Global-sample SBFN members, by SBFN region.

Region	Member economy-years
Latin America and Caribbean	17
Africa	12
Europe and Central Asia	11
East Asia and Pacific	11
South Asia	6
Middle East and North Africa	4
Total	61

1213 **Additional material**

1214 *Online appendix*

1215 Appendix A, Appendix B, and Appendix C appear above; per-country
1216 source citations at the level of detail compiled during data construction (in-
1217 dividual SBFN Global Progress Report page references, country-addendum
1218 URLs, and IFC accession-announcement links for every non-obvious deter-
1219 mination in the primary sample; the full 162-row global-sample SBFN/WGI
1220 table underlying Appendix C) are available in the online supplementary
1221 material accompanying this submission.

1222 *Data availability*

1223 Data in support of the findings of this study is available from the corre-
1224 sponding author upon reasonable request, subject to the World Bank Enter-
1225 prise Surveys' own terms of use for the underlying firm-level microdata (enter-
1226 prisesurveys.org, free registration required), including the standardized cross-
1227 country database and Productivity Database underlying the global sample.
1228 The country-level SBFN policy-status (including the SBFN data portal's
1229 member roster underlying the global sample) and Worldwide Governance In-
1230 dicators datasets constructed for this paper, the country-year World Develop-
1231 ment Indicators panel and Callaway-Sant'Anna event-study code underlying
1232 Section 6 (all freely redistributable, no registration required), together with
1233 the full analysis and replication code, are available in a public GitHub repos-
1234 itory; the repository link is withheld from this anonymized manuscript, and
1235 repository access is restricted for the duration of double-anonymized review,
1236 to avoid identifying the authors to reviewers. The link will be provided to

1237 the editor upon submission and made publicly accessible again to readers
1238 upon acceptance.

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